

Response to the Editor and Reviewers

Climate outweighs human effects on pollen assemblage dynamics from 8.5 to 2.0 ka BP

EDITOR'S REQUESTS

OVERVIEW OF THE REVISION

Authors We thank the Editor and reviewers for their careful assessment of our manuscript. Since receiving their comments, we have undertaken a substantial analytical and textual revision. We strengthened the central H1 comparison at both scales: the spatial analysis now accounts for time within each pollen sequence, while the temporal analysis accounts for spatial structure among sequences within each continental-group time slice. Following revisions to the model specifications, we refitted the complete set of hierarchical generalised additive models (hGAMs) used to describe temporal trajectories of pollen assemblage properties and predictor histories, then regenerated their predictions and figures. We corrected the extraction of hierarchical variation-partitioning contributions and now report overall model fit separately from the relative contributions of climate and human activity.

Authors We added quality-control and sensitivity analyses of pollen-count coverage, correlations among pollen assemblage properties, shared and negative contributions, residual spatial dependence, spatial thinning, omission of geographic groups, the archaeological radiocarbon neighbourhood, and the definition of the human-activity proxy. After adding these checks and sensitivity analyses, we continue to find the same general pattern as in the original H1 analysis: climate had the larger estimated contribution than human activity at the broad spatial and temporal scales examined.

Authors We have revised the Methods, Results, Discussion, figures, and Supplementary Information to make these changes and their limitations clear. The responses below address the editorial requirements and each reviewer comment in detail.

In particular, the following editorial requirements need to be met:

- Provide a compelling case that climate outweighs human effects on vegetation assemblage dynamics in the Holocene, including a thorough analysis of the results, clearly indicating the variance explained by models.

Response. We have strengthened the evidence for the central conclusion and clarified what the models do and do not show. The revised article now distinguishes the overall explanatory power of each model from the relative contributions associated with palaeoclimate and human activity. During revision, we also corrected the extraction of these contributions and examined whether the conclusion depended on alternative treatments of shared or negative components. Palaeoclimate had the larger estimated contribution in 891 of the 1,098 pollen sequences in the controlled spatial analysis (81.1%). For these same sequences, total adjusted R^2 was 0.440 [0.426, 0.449] (mean, 95% CI) and 0.450 [0.326, 0.567] (median, interquartile range). The regional analysis through time gave the same general result: climate had the larger estimated contribution in 69 of 70 comparisons for which both climate and archaeological evidence could be evaluated.

Changes in the revised article. These results are reported in “H1: Holocene human impact secondary to palaeoclimate” and in Figs. 3–4. The Supplementary Information documents model fit, uncertainty arising from shared explanatory variation, and alternative ways of summarising the contributions. The Discussion now makes clear that these contributions divide the variation explained by a model; they do not represent proportions of all ecological change recorded in the pollen data.

- Justify the usage of data from multiple scales and fully discuss the limitations of your methods, particularly as raised by reviewer #2.

Response. We have revised the article to describe explicitly the different spatial worlds represented by the evidence. A pollen sequence reflects vegetation within a source area that differs among lakes, bogs, and depositional settings. The climatic values describe conditions at the location of each sequence, whereas the archaeological SPDs draw on radiocarbon dates from a much broader surrounding landscape. The expert chronologies represent regionally informed interpretations of changes within individual pollen diagrams. These sources cannot be made spatially equivalent within a global compilation. We therefore restrict our conclusion to continental-group and regional patterns between 8.5 and 2.0 ka BP and discuss how scale affects ecological interpretation. We also compare results obtained with 250 and 500 km archaeological neighbourhoods and with alternative evidence for human activity. The revised Discussion gives fuller attention to pollen production and count size, regional differences in archaeological knowledge, climatic smoothing, shifting Holocene climates, and the persistence of spatial and temporal structure in ecological archives.

Changes in the revised article. The spatial meaning and limitations of each source are now explained in the Methods sections on data filtering, spatial units, human activity, palaeoclimate, and H1. Their consequences for interpreting local and regional ecological change are developed in the Discussion, with the supporting analyses reported in the Supplementary Information.

- Clearly explain your methods in detail.

Response. We have expanded the Methods so that readers can follow the evidence from pollen counts and archaeological dates to the ecological comparison presented in the Results. The revised text explains how sequences were selected, how the ten pollen assemblage properties were calculated, how temporal trajectories were summarised without allowing long or densely sampled sequences to dominate, and how the climate and human-history variables were assembled. It also distinguishes the amount of variation explained by a model from the division of that explained variation between climate and human activity. The treatment of time, spatial pattern, and the supplementary robustness analyses is now stated explicitly. The guide below shows which pollen sequences or continental-group time intervals contribute to each part of the study.

Changes in the revised article. These explanations are integrated throughout the Methods. More specialised checks are described under “Result-validation analyses” in Methods, and each figure and table now states the sequences, regions, or time intervals on which it is based.

Analysis subset guide (authors’ summary).

The rows below refer to different bodies of evidence and, in two cases, different units of observation. They are therefore not successive losses from a single dataset. Full definitions are provided in Supplementary Table S1.

Evidence used	Count	Sequences or comparisons represented
Prepared pollen collection	1,270	Pollen sequences after baseline preparation
Sequences considered for the initial spatial analysis	1,262	Initial set before model eligibility checks
Sequences with an estimable initial comparison	1,156	Sequences for which climate and human contributions could be estimated
Principal spatial analysis	1,098	Pollen sequences analysed after accounting for time within each sequence
Principal regional analysis through time	70	Regional time intervals analysed after accounting for geographical pattern

Spatial comparison of
archaeological and expert
evidence

151 Sequences with both
archaeological and expert
evidence available for
comparison

Temporal comparison of
archaeological and expert
evidence

61 Regional time intervals
with both forms of
human evidence available

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author)

Review of the manuscript

“Climate outweighs human effects on vegetation assemblage dynamics in the Holocene” by Felde et al.

The manuscript submitted by Felde et al. presents a substantial contribution to the field of paleoecology. The study is based on a large dataset of pollen records spanning all continents to investigate the relative influence of climate and human activity on vegetation compositional changes throughout the Holocene. The authors use modelled climate data and archaeological radiocarbon dates, along with expert knowledge on human activities derived from pollen records, to disentangle the effects of climate and human activities on vegetation/pollen composition and diversity. The analytical framework is well developed, employing state-of-the-art statistical methods to explore large-scale pollen composition and diversity changes. The use of Pollen Assemblage Properties (PAPs) to assess compositional changes is innovative and adds a new dimension to paleoecological interpretation. Overall, this is an important and ambitious manuscript that I highly recommend for publication.

My major comments on the manuscript are listed below.

Response. We thank the reviewer for the constructive assessment. The revision defines more carefully the ecological meaning of the pollen assemblage properties, the historical interval and spatial scales to which the conclusions apply, and the strengths and limitations of the climate and archaeological evidence.

Scope and framing of the study period

The title suggests an analysis of the entire Holocene, yet the dataset ends approximately 2000 years ago. This period excludes the last two millennia, during which anthropogenic impacts on vegetation intensified, including major deforestation events and agricultural expansion. The title should be revised to more accurately reflect the temporal scope of the data and findings.

Furthermore, some statements in the discussion, such as the caution against the narrative that human transformation is always destructive, may appear strong given the temporal cut-off at 2 ka BP. I suggest to soften or contextualize this statement to avoid misinterpretation.

Response. We agree. The title now specifies the interval 8.5–2.0 ka BP and refers to pollen assemblage dynamics. This is the period for which the archaeological SPD evidence supports the principal comparison. Some expert-identified events occur after 2.0 ka BP, but these younger observations are shown separately and do not extend the principal conclusion into the last two millennia. We have also removed the broad statement that human transformation was not destructive. The revised Discussion instead emphasises that our result concerns the spatially aggregated ecological patterns represented in this study and does not diminish the extensive historical and palaeoecological evidence for substantial local human transformation.

Changes in the revised article. The title is now “Climate outweighs human effects on pollen assemblage dynamics from 8.5 to 2.0 ka BP.” The same temporal and ecological scope is maintained in the Abstract, hypotheses, Results, Discussion, and Conclusion. Younger event observations are identified separately in Fig. 4.

Use and justification of climate model outputs and human activity proxies

While the use of paleoclimate model data is justified by the limited spatial coverage of proxy-based climate reconstructions, a brief discussion of the benefits and limitations of using model outputs (e.g., smoothed trends and difficulties to simulate climate in some regions) should be included. This will help the readers to better understand the implications using climate model outputs.

Response. We agree and have added a fuller discussion. CHELSA-TraCE21k provides a spatially continuous and internally consistent description of climatic change across regions where proxy reconstructions remain uneven. This is valuable for a global comparison, but the simulations smooth short-lived and local climatic variation, depend on model assumptions, and may not reproduce conditions within the actual pollen source area, especially where elevation or topography changes over short distances. We therefore interpret the climate variables as broad representations of climatic history rather than exact reconstructions of the environments surrounding individual sites.

Changes in the revised article. The advantages and limitations of the simulated climate histories are now explained under “Palaeoclimate” and revisited in the Discussion when comparing local ecological change with broad climatic patterns.

In a same way, the use of summed probability densities (SPDs) from archaeological radiocarbon dates is appropriate but should be more critically discussed, although the authors already mention some of the issues. These can underestimate human presence, especially in regions or

periods with sparse data. Furthermore, the expert-derived “human-impact events” from pollen data introduce a degree of circular reasoning as it is based on pollen information. The authors should mention the pros and cons of using such approach.

Response. We agree. We retain archaeological SPDs as validated proxies for regional human population changes, while making clear that they are histories of dated archaeological activity rather than direct censuses of past populations. Their form is influenced by preservation, calibration, archaeological research traditions, and uneven knowledge among regions and periods. SPD remains the principal human-history proxy because it is independent of the pollen response and can be compared across much of the study area. The expert-derived events contribute a different kind of evidence: regionally informed indications of land use and vegetation disturbance within pollen diagrams. They can reveal changes not represented by a regional SPD, but they are unevenly available and partly circular because both the event and the ecological response may be inferred from the same pollen sequence. We therefore use them to examine sensitivity to the choice of historical evidence, not as independent confirmation.

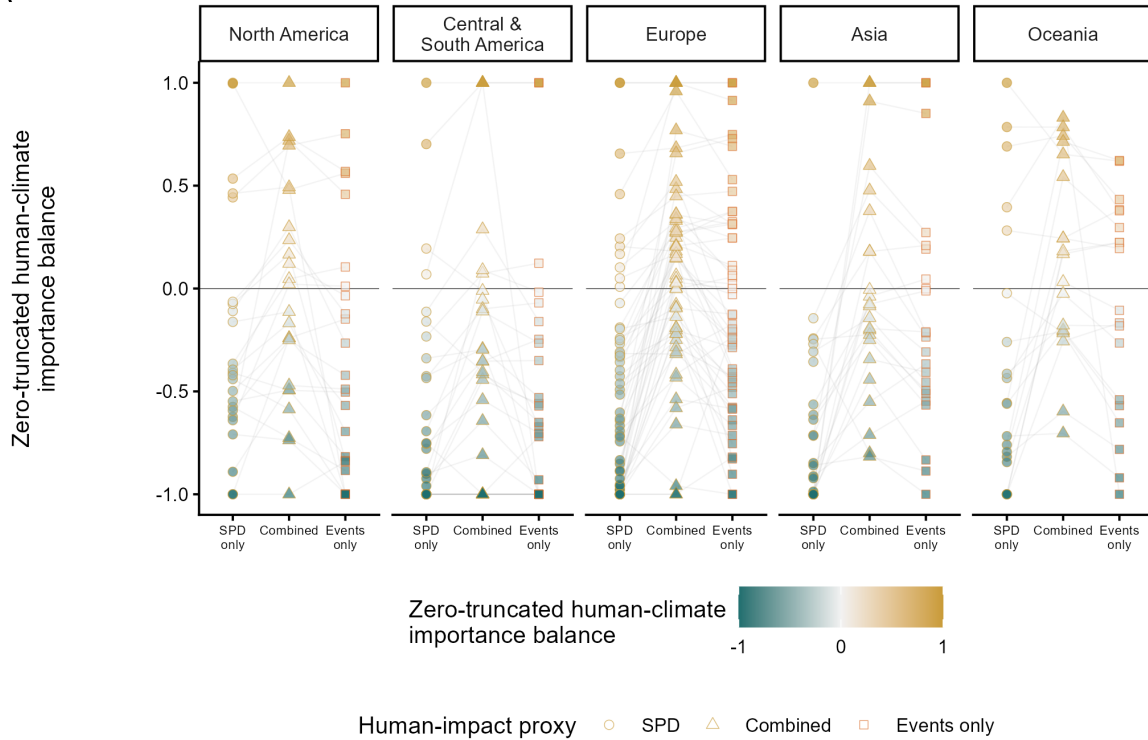
We also repeated H1 within the subset with observed event chronologies, representing human activity in turn by the archaeological SPD alone, the expert-derived events alone, and both together. Across 151 spatial sequences and 61 continental-group–age comparisons, climate retained the larger median estimated contribution under each definition (Supplementary Figs. S9 and S39). This preserves the broad direction of the principal SPD analysis, although the combined spatial balance was close to parity and estimates for individual sequences and time intervals changed. Supplementary Fig. S40 shows the temporal changes relative to SPD alone. Because the events are partly inferred from pollen diagrams, these repetitions test sensitivity to the human-activity proxy rather than provide independent confirmation.

Changes in the revised article. These complementary strengths and limitations are now explained in the Methods and Discussion. Supplementary Figs. S9, S39, and S40 show the SPD-only, event-only, and combined comparisons and the temporal changes relative to SPD alone.

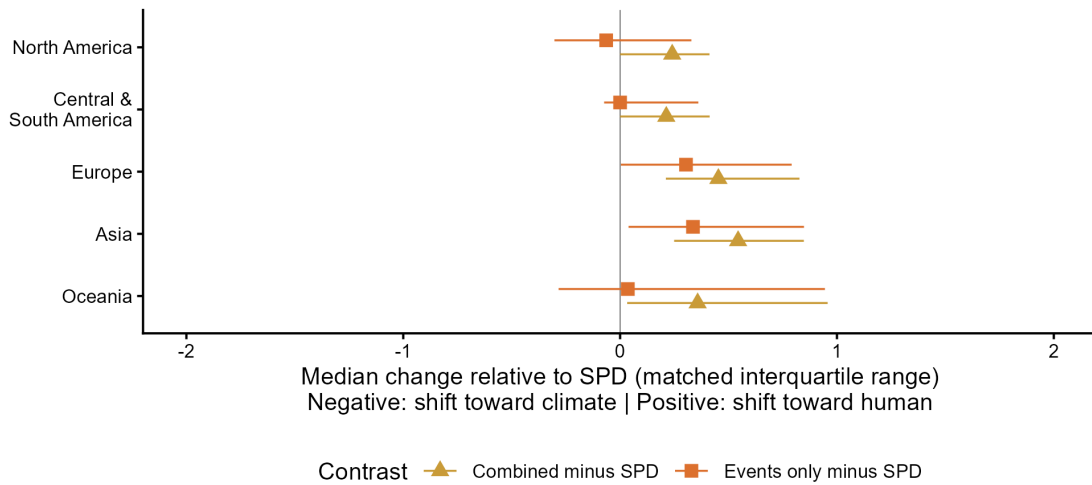
For direct comparison, we reproduce the spatial and temporal proxy profiles and the matched temporal changes below.

Regional-event sensitivity of spatial human-climate balance

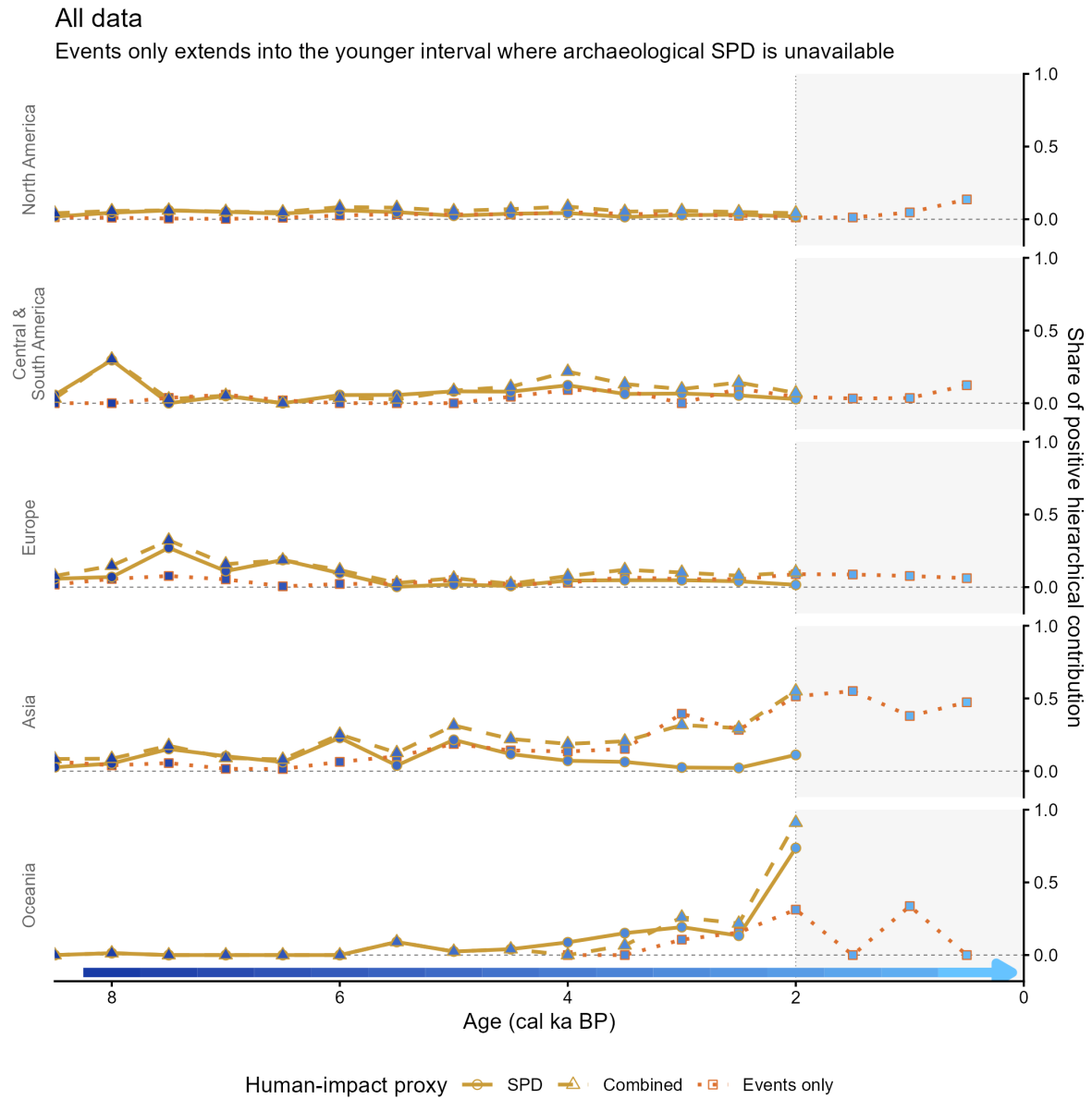
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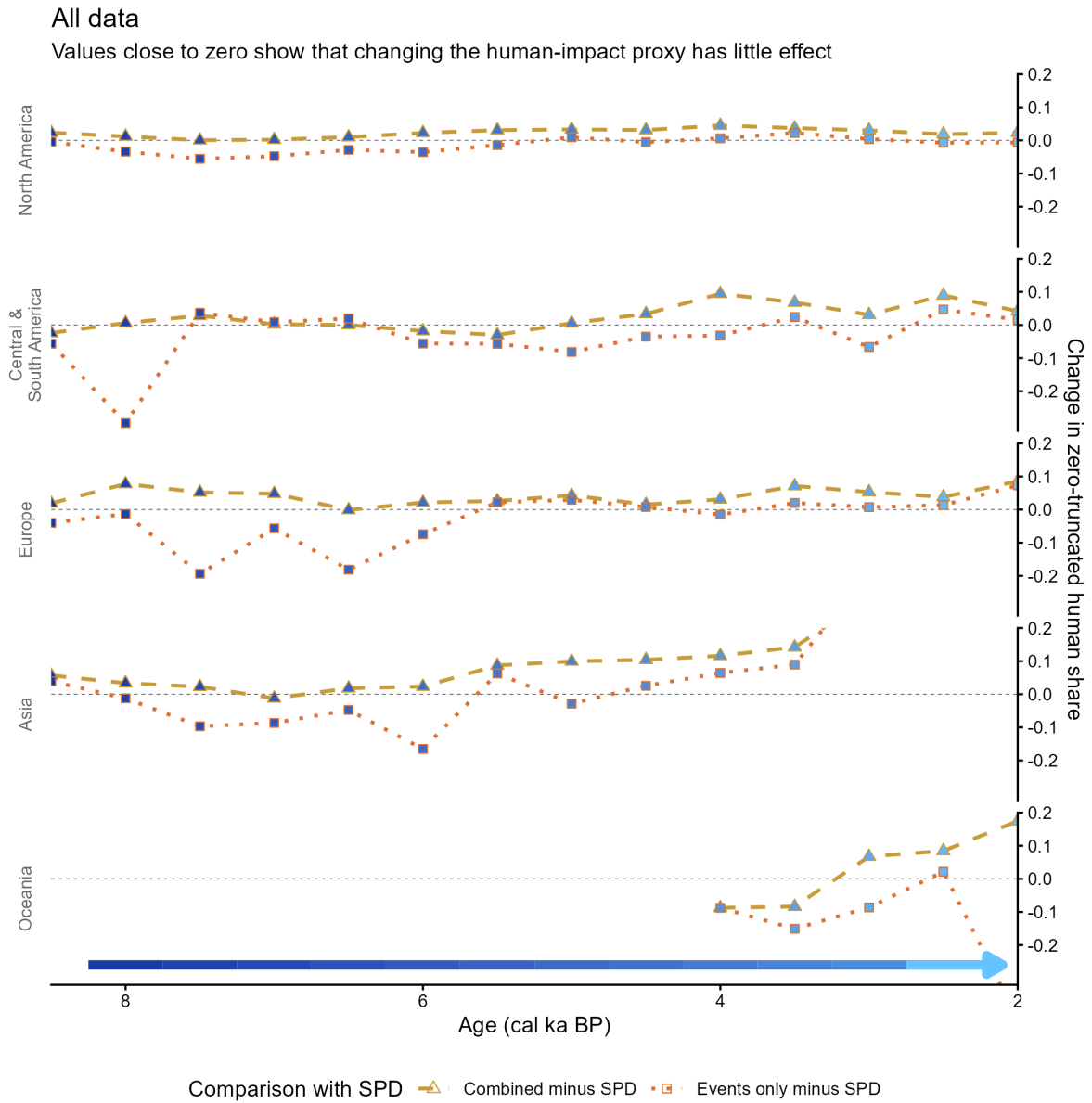
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Supplementary Fig. S9 (reproduced). Spatial human-event sensitivity. The three human-proxy definitions are compared among sequences with observed event chronologies. Negative balances favour climate; the combined estimate moves closer to parity for some sequences.



Supplementary Fig. S39 (reproduced). Temporal human-event profiles. The spatially controlled analysis shows all available continental-group time slices. Event-only profiles extend into younger intervals where the archaeological SPD is unavailable; those intervals are outside the matched three-way comparison.



Supplementary Fig. S40 (reproduced). Temporal human-event changes. Within matched continental-group–age comparisons, the lines show changes in the human allocation relative to SPD-only. Positive values shift toward greater human allocation; negative values shift toward climate.

Background information and previous studies

The introduction would benefit from presenting a bit more prior studies that used similar approaches, e.g., Kuosmanen et al. (2018), Marquer et al. (2017), and Reitalu et al. (2015). These studies provide an important context for methods, data types, and interpretations of vegetation change drivers, although for Europe and regional scales only. Their inclusion would strengthen the rationale for the current study.

Response. We agree that the regional literature is essential context. The Discussion now relates our study to European and other regional attempts to separate climatic and human influences on past vegetation. We also clarify an important ecological distinction: our response variables describe richness, evenness, turnover, and other properties of pollen assemblages, whereas several earlier studies reconstructed changes in particular taxa or vegetation composition.

Changes in the revised article. The revised text discusses Kuosmanen et al. (2018), Marquer et al. (2017), Reitalu et al. (2013) on climatic and human influences on forest composition, and Reitalu et al. (2015) on functional and phylogenetic diversity inferred from pollen.

Interpretation of hypotheses and results

Hypothesis 1 states that if human impact is an important driver, its influence should match or exceed that of climate. I would say “the major driver” rather than “important driver” in the case it exceeds climate.

Response. We agree and now describe human activity as the proposed “primary driver” in H1. This wording reflects the intended comparison with climate without implying that a smaller estimated contribution would make human influence ecologically unimportant.

Changes in the revised article. H1 now reads: “The first hypothesis (H1) proposed that human impact was the primary driver of changes in PAPs at continental-group and regional scales between 8.5 and 2.0 ka BP.”

The authors conclude that climate dominates vegetation change at large spatial scales rather than local scales. This aligns with prior studies. I agree with the authors and clearly support the fact that at finer spatial resolutions, human impacts become more visible and potentially dominant. It is important to mention that the use of PAPs may homogenize vegetation signals and amplify climate-related patterns. This should be acknowledged.

Response. We appreciate this concern, but do not assume that using PAPs inherently amplifies climatic patterns. PAPs describe different properties of the pollen assemblage in each individual sequence; calculating them does not average sequences across a region. Rich-

ness, evenness, turnover, and rate of change can reveal assemblage reorganisation without relying on the response of a single pollen type, although a change confined to particular taxa may leave some of these properties largely unchanged. In the spatial H1 analysis, climatic and human contributions are estimated within each sequence, with time controlled, before those estimates are summarised across continental groups and regions. The human proxy had the larger estimated contribution in 207 of 1,098 controlled sequences (Fig. 3). This minority of human-favoured sequences shows that the sequence-level analysis does not impose a uniform climatic result, although neither the PAPs nor the human proxies identify which local practice caused a particular change. A broad geographical summary can give less prominence to heterogeneous local histories, and we now make that limitation explicit. The reduced-PAP sensitivity retains the aggregate climate-over-human ordering (Supplementary Fig. S30), but a taxon-level comparison would be needed to test whether response choice systematically favours either predictor.

Changes in the revised article. The Discussion now distinguishes sequence-level fitting from geographical summaries, explains which aspects of an assemblage the PAPs capture, and cautions against using the broad climatic pattern to discount well-supported local histories of human transformation.

For direct assessment, we reproduce Fig. 3 and the reduced-PAP sensitivity shown in Supplementary Fig. S30.

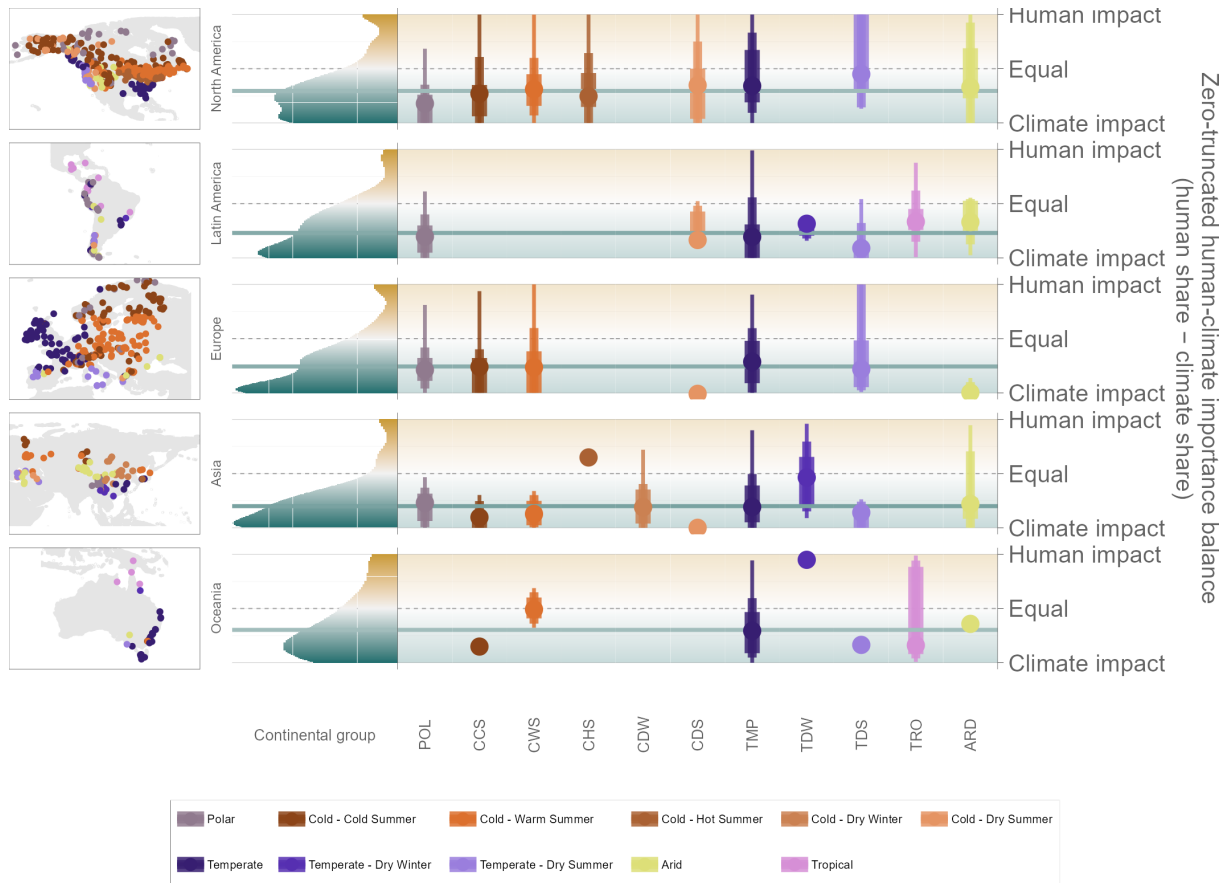
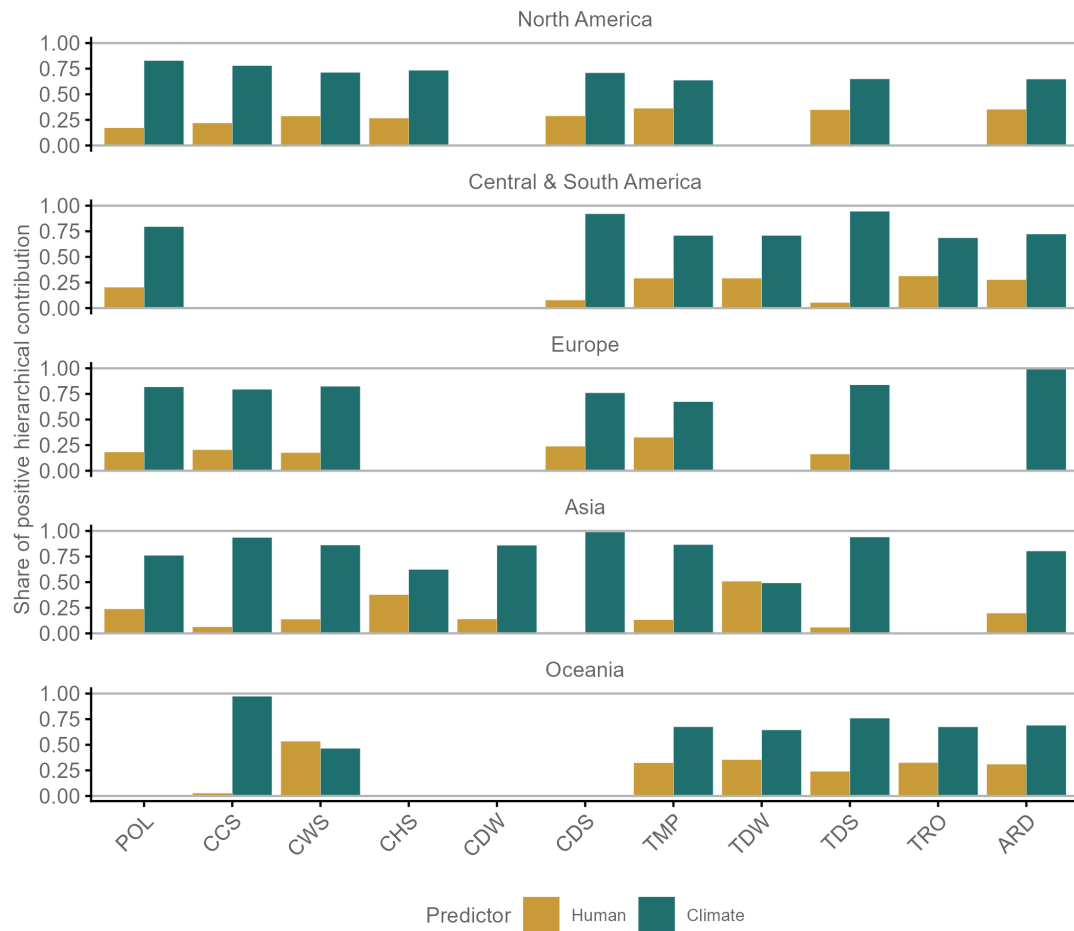


Fig. 3 (reproduced). Spatial H1 after controlling for time. Sequence-level human-minus-climate balances are shown alongside regional and continental-group summaries. Positive balances favour the human proxy; negative balances favour climate. Displayed balances are zero-truncated, so a value of zero does not establish that a driver had no ecological effect.



Supplementary Fig. S30 (reproduced). Reduced-PAP response sensitivity. The spatial H1 analysis was repeated with a less redundant set of PAPs. The broad climate-over-human ordering persists in this zero-truncated display, although estimates for individual sequences can change.

The manuscript would be greatly improved if the authors provide a discussion of vegetation changes themselves, and how specific taxa or assemblage types respond to climatic or anthropogenic drivers. Such discussion would help ecological interpretations.

Response. We agree that the ecological context of the PAPs is important. Our question, however, is how strongly palaeoclimate and validated proxies for regional human population changes are associated with multiple properties of pollen assemblages, rather than how particular taxa or vegetation types changed through time. Richness, evenness, turnover, and rate of change describe different dimensions of assemblage change, but they cannot establish which taxa produced that change or attribute a taxon's response to a specific driver. A discussion of such responses based on our results would therefore go beyond what the analyses can support. Complementary large-scale pollen studies examine European floristic patterns^{1,2}, North American woody pollen diversity³, Asian assemblage properties⁴, and Holocene floristic diversity across continents⁵. The more recent European study also examines changes in human-associated pollen taxa². We now point readers to this literature while retaining the distinction between the ecological properties estimated here and taxon-specific vegetation histories.

Changes in the revised article. The Discussion explains the ecological meaning of the PAPs, cites the complementary regional and cross-continental syntheses, and states that attribution of changes in individual taxa would require a separate analysis.

Data representativeness and site characteristics

The choice of 1262 pollen sequences is impressive, but more information is needed about the selection criteria. I would have assumed that much more sites would be available at such spatial scales.

Response. We have expanded the account of how the pollen archive was assembled. Duplicate sequences were removed; samples outside the study interval or lacking adequate chronological control were excluded; each retained sample contained at least 25 terrestrial pollen grains; and each sequence contained at least five usable samples. Taxonomic names were harmonised before the assemblage properties were calculated. Africa was not included because, at the time the compilation was assembled, the available sequences were too sparse when the compilation was assembled to support comparable regional analyses. This procedure yielded 1,270 prepared sequences, of which 1,262 entered the original H1 spatial analysis. Particular analyses use fewer sequences where the necessary climate, archaeological, ecological, or control information is absent.

Changes in the revised article. The full selection criteria are now listed under "Data filtering." Supplementary Table S1 and the guide at the beginning of this letter identify the sequences or regional comparisons represented in each analysis. Because the original search

did not retain a complete count of every potentially available public sequence, we do not present the prepared collection as a fully quantified global census of pollen sequences.

Site characteristics (e.g., basin size, proximity to human settlements) can significantly influence pollen source area and the sensitivity to land use signals. This aspect, although mentioned by the authors, should be discussed a bit more in details, especially when interpreting local vs regional patterns.

Response. We agree. Basin size and depositional setting determine how much of the surrounding vegetation contributes pollen to a sequence, while distance from settlements or cultivated land affects the likelihood that local land use will be recorded. These differences are ecologically important when a site-scale pollen signal is compared with an archaeological history assembled over hundreds of kilometres. However, consistent information on basin geometry, depositional environment, and settlement proximity is unavailable for much of the global compilation, so these factors could not be represented uniformly in the models. We consequently restrict the interpretation to broad spatial patterns and discuss why individual sites may depart from them.

Changes in the revised article. The Methods and Discussion now give greater attention to basin setting, pollen source area, and the contrast between local land-use histories and broad archaeological neighbourhoods. A separate comparison shows how the results change when dates are drawn from 250 rather than 500 km around a pollen site.

The analysis assumes stable spatial climate zones. However, climate zones have shifted over the Holocene, which could affect interpretations of vegetation-climate relationships. A brief discussion of this dynamic would be valuable.

Response. We agree that climatic boundaries shifted during the Holocene, but our H1 analyses do not assume that present-day Köppen–Geiger classes persisted through time. The classes were used to assign sequences to smaller *regions* within the five *continental groups*, making heterogeneous results easier to present and compare; they were not used as reconstructions of past climate. In spatial H1, human and climate contributions were estimated within each individual pollen sequence before the estimates were summarised by region or continental group. In temporal H1, the models compare sequences within 500-year continental-group time slices, with spatial structure controlled, rather than treating regions as fixed climate zones. The palaeoclimate predictors vary through time at the locations of the sequences. Regions do serve as analysis units in H2, so its regional comparisons are conditional on this pragmatic grouping. Fig. 2 displays distinct PAP trajectories among the regional groups, based on sequence-level predictions; these descriptive summaries are not inputs to H1. Fig. 3, reproduced above, maps the eligible sequences and shows the distribution of their individual spatial H1 balances. Neither figure requires the region assigned from modern Köppen–Geiger classes to have retained the same climate throughout

the Holocene.

Changes in the revised article. The Methods now state where regional assignments enter the analyses and distinguish them from the time-varying palaeoclimate predictors. The Discussion retains the caution that regional trajectories are summaries for practical groups, not histories of fixed ecological units.

For direct assessment, we reproduce Fig. 2 below; Fig. 3 is reproduced above with the preceding response.

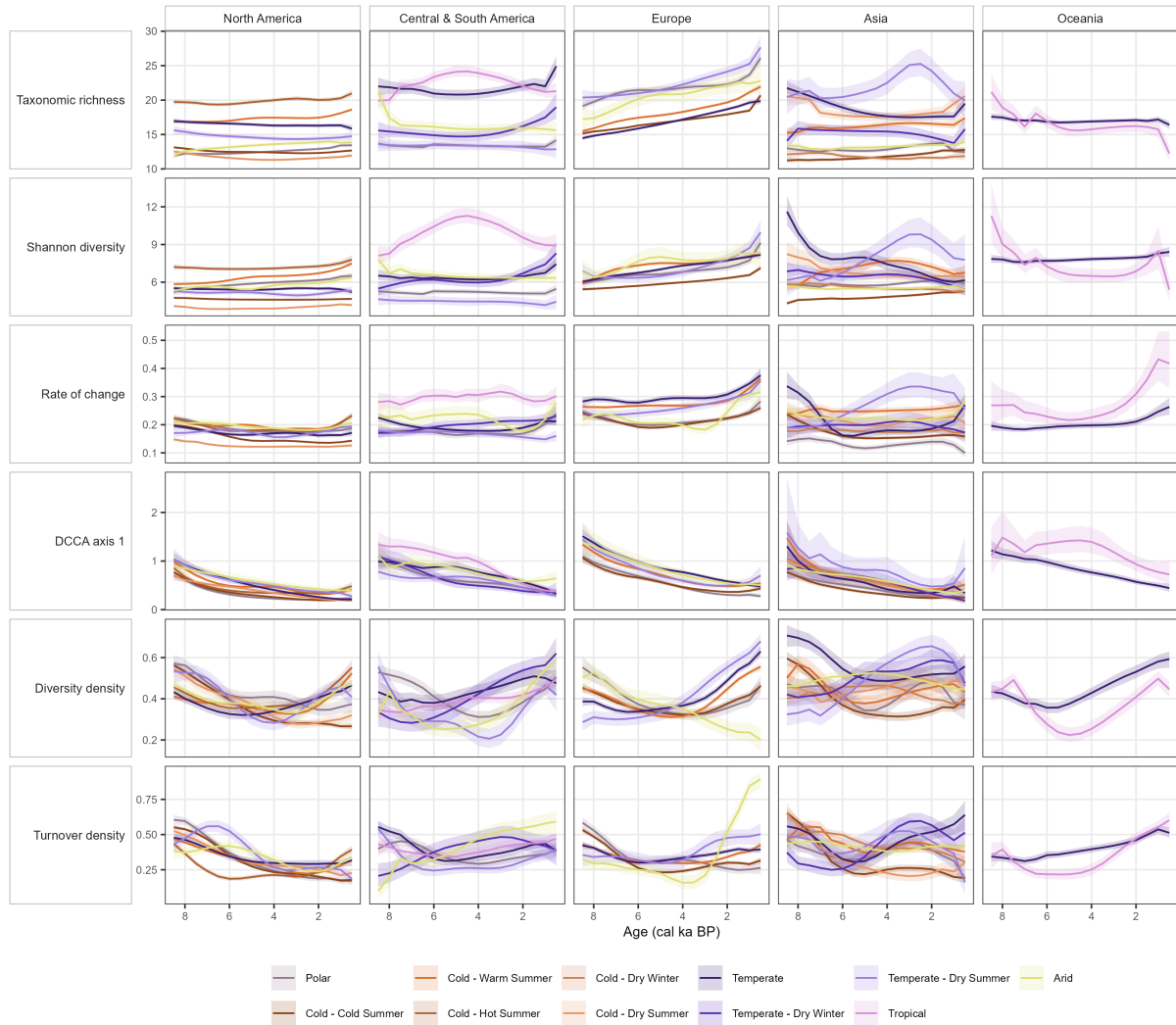


Fig. 2 (reproduced). Temporal trajectories of six PAPs. Each regional curve is an equal-weighted average of sequence-level posterior predictions within the observed coverage of its sequences. These descriptive trajectories display differences among the practical regional groups and do not enter the H1 models.

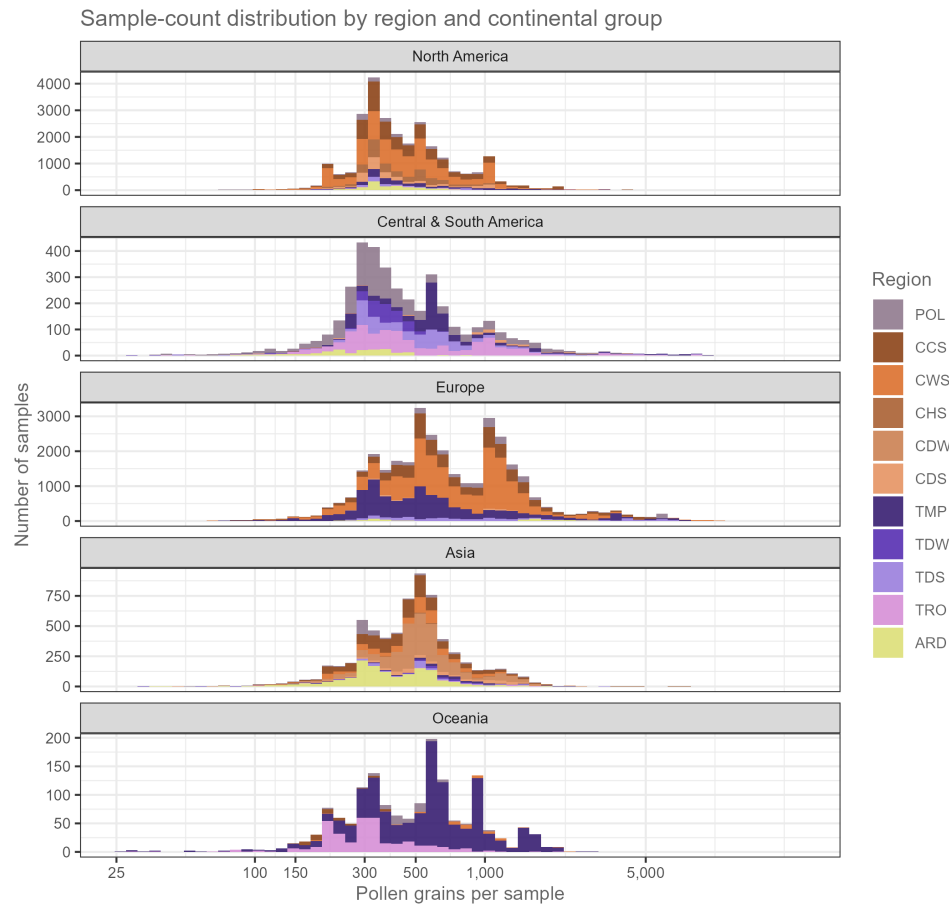
Thresholds for pollen counts

The decision to lower the minimum pollen count threshold from 150 to 25 grains per sample raises some questions about data quality. A threshold as low as 25 may introduce significant noise and reduce the reliability of PAPs. The rationale for this decision should be clearly explained and its implications discussed.

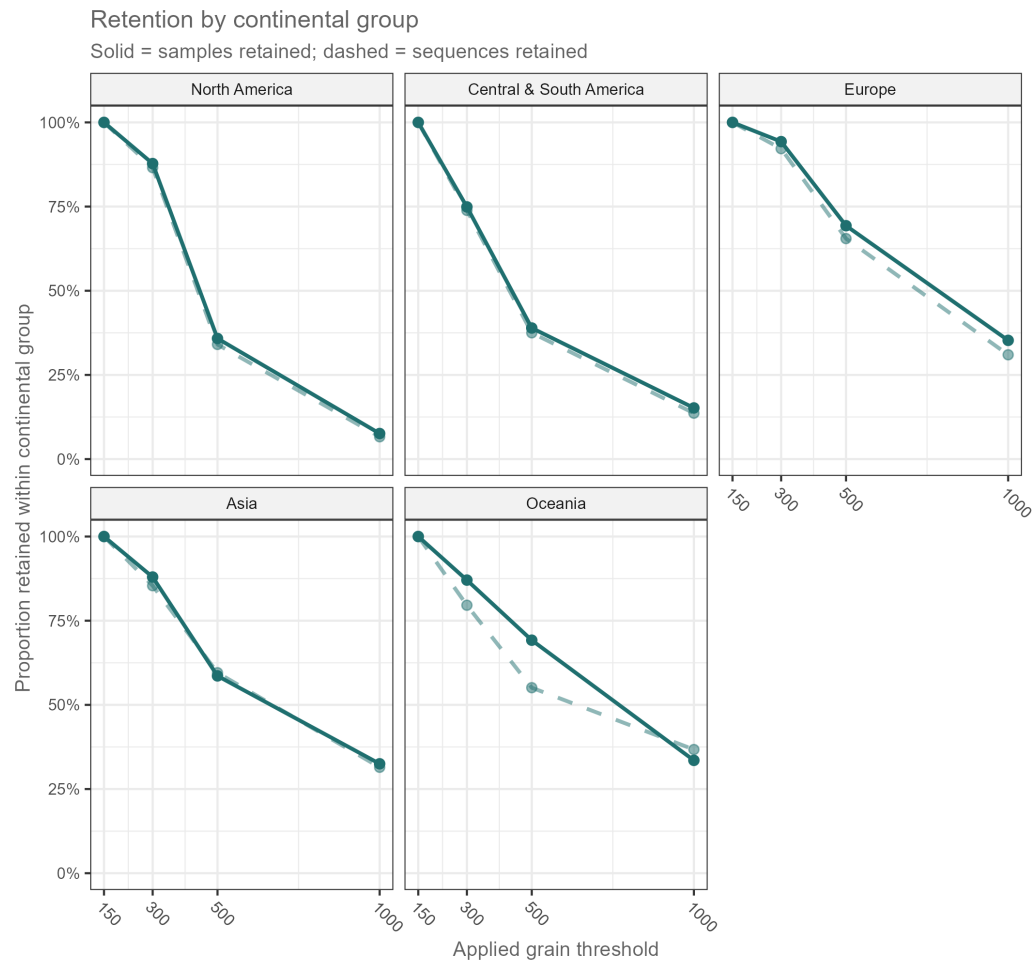
Response. We agree that assemblage properties estimated from very small pollen counts are less precise, but 25 grains is the lower limit for retaining an individual sample, not a representative count in the analysis. In the retained count-based archive, only 84 of 72,191 samples (0.12%) had 25–49 terrestrial pollen grains; 98.7% had at least 150 grains (Supplementary Fig. S23A). A retained count-based sequence must have more than half of its samples at or above 150 grains, the usual rarefaction target; a lower sequence-specific target can be used when some of its retained samples fall below that count. Supplementary Fig. S23B shows the cost of stricter thresholds: a 300-grain target retains 1105 of 1270 sequences, a 500-grain target retains 616, and a 1,000-grain target retains 246. Losses differ among continental groups and periods, so demanding larger counts would reduce the geographical and temporal breadth of the evidence. These comparisons increase the rarefaction target and its associated sample minimum; they measure coverage, not the effect of refitting H1 at those thresholds. We retain the lower bound to preserve sparsely represented sequences while acknowledging that estimates based on small samples carry greater uncertainty.

Changes in the revised article. The Methods now distinguish the rare 25–49-grain samples from the usual counts, explain the sequence-retention and rarefaction rules, and point to the count distribution and retention diagnostics in Supplementary Fig. S23. Supplementary Table S3 reports the losses under stricter thresholds, and the Discussion addresses finite counts and pollen production as limits on ecological interpretation.

For direct assessment, we reproduce both panels of Supplementary Fig. S23.



Supplementary Fig. S23A (reproduced). Sample-count distribution. Terrestrial pollen grains per retained count-based sample, displayed by continental group and region. Percentage-only sequences are excluded. Very few count-based samples lie near the 25-grain lower limit.



Supplementary Fig. S23B (reproduced). Retention under stricter targets. Dashed lines show retained sequences and solid lines retained samples as rarefaction targets rise. These are coverage diagnostics, not H1 refits.

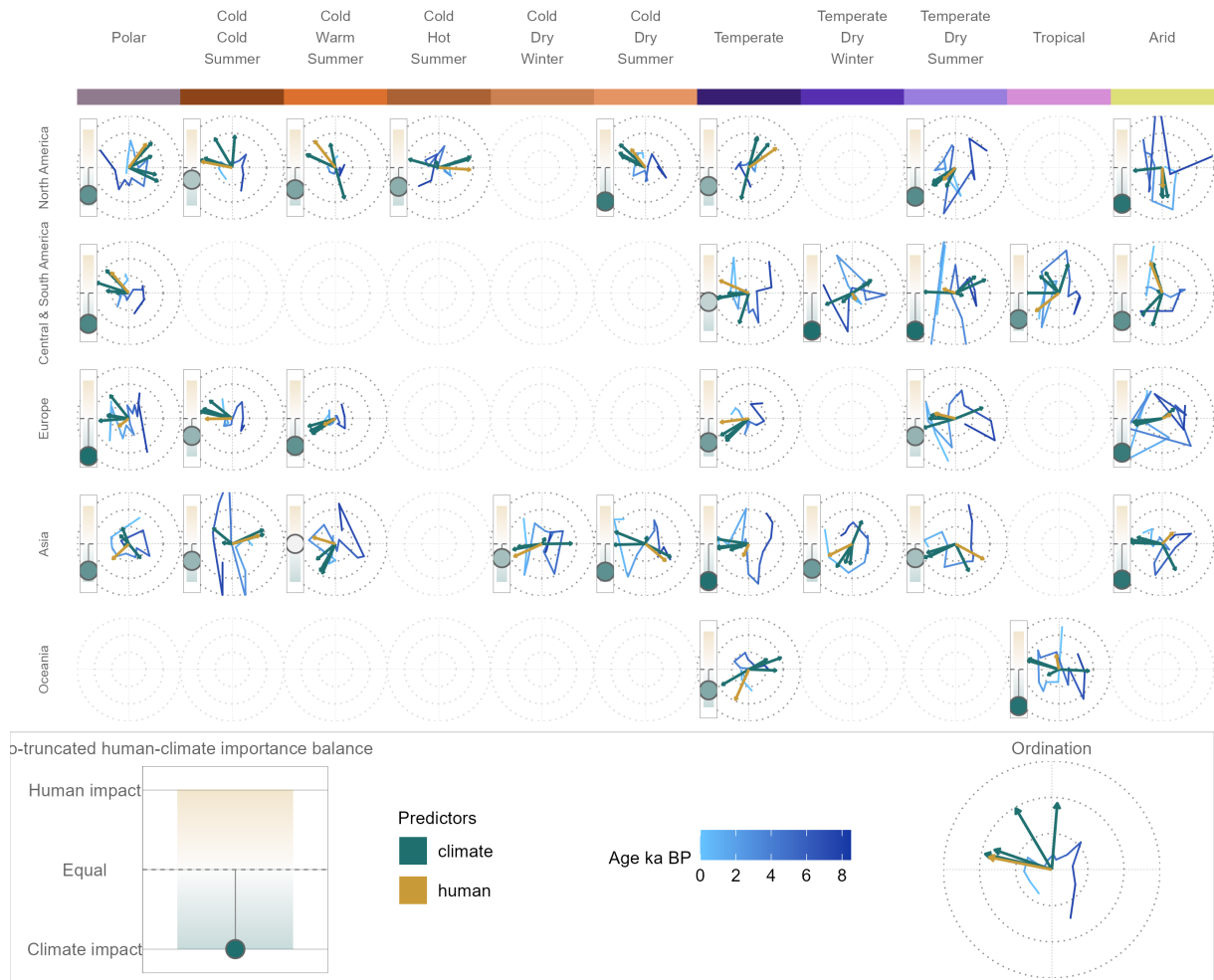
Figures

Figure 4 shows a lot of information, and the plots are too small. It is difficult to see the major patterns. Furthermore, the panel labels in Figure 2 appear to be misaligned (E and D).

Response. We agree that the submitted figures were crowded and that the panel labels were difficult to follow. We reorganised the main figures so that the study design, descriptive PAP trajectories, spatial predictor comparison, and temporal predictor comparison are presented separately (Figs. 1–4). The misaligned panel lettering of the submitted Fig. 2 is absent from the revised display; the current figures use labelled rows and columns. We also redesigned the former Fig. 4, now Supplementary Fig. S10, rather than simply moving it: its ordination panels have more space, the human–climate balance is shown beside each panel, and the explanatory key distinguishes the age trajectory from the predictor arrows and balance. All ten PAP trajectories and the additional robustness displays remain available in the Supplementary Information.

Changes in the revised article. The main figures now separate the principal questions, and the redesigned Supplementary Fig. S10 presents the H2 interrelationships at a larger, more readable scale with a clearer key and caption.

For direct assessment of the former Fig. 4, we reproduce its revised version below.



Supplementary Fig. S10 (reproduced). H2 PAP interrelationships. Revised ordinations and adjacent human–climate balances by continental group and region. Blue trajectories run from older to younger time slices; gold and teal arrows indicate associations with the human and climate predictors. Blank panels lack an estimable result.

This manuscript provides a novel and valuable analysis of Holocene vegetation dynamics at broad spatial scales. The conclusions are well supported, and my major comments focus on some aspects of the methods and interpretation that require clarification and additional nuance. By addressing these issues, the manuscript will be greatly improved. Again, I recommend publication after these revisions are done.

Response. We thank the reviewer again. The revised article now places the statistical comparison within a clearer account of the pollen archive, the historical proxies, the spatial scale of ecological change, and the limits of broad-scale inference.

Reviewer #2 (Remarks to the Author)

This paper uses a global dataset of fossil pollen, paleoclimate, archaeologically-attested radiocarbon dates and human indicator pollen types to investigate the relative importance of humans and climate on aspects of pollen diversity and compositional changes from 8.5 cal k BP onwards. Previous work has described changes to various aspects of pollen diversity and compositional turnover measures over multi-millennial timescales across much of the earth surface. Other studies have investigated the relative strengths of relationships between humans versus climate on pollen diversity and compositional turnover for different continents and regions. However, this study represents the first attempt to quantify the relative importance of humans and climate on pollen diversity and compositional turnover measures at a global scale in the Holocene. Using a new method to the field of paleoecology, the authors claim that, at continental and regional scales, climate was the primary driver of pollen diversity and compositional turnover measures over the period 8 - 2.5 cal k BP. This work adds a novel dimension to the conversation (a direct comparison of the strength of drivers globally) and thus represents an interesting and worthwhile contribution to the literature.

Response. We thank the reviewer for recognising the scope and novelty of the study.

I do, however, have a number of data presentation and methodological concerns that I would need to see addressed to have confidence in the results presented here. I also see no advantage of including the human event variable here. I understand that the authors would prefer to be able to run these analyses up to the present, but I find the variable reductive and without nuance, it is un-tested and it is much too coarse for it to ever be likely to capture changes to local PAPs. In addition to methodological considerations, I would also like to see an adjustment to the title. In its current form, at least to me, the title suggests that this is a study encompassing the full Holocene, though the primary analyses span the period 8.5 - 2 cal k BP. This is very important because recent work has shown substantial increases in multiple measures of pollen diversity from 2.5 cal k BP onwards across the planet (e.g. in Europe, Giesecke et al, 2019; North America, Stegner and Spanbauer, 2023; global, Mottl et al, 2021 and Gordon et al, 2024), a period of known rapid and major change in human population size, culture, etc. These changes are missed in these analyses given the cut-off date. Climate outweighs human effects on vegetation

assemblage dynamics 8 - 2.5 ka BP (or similar) would seem to me to be a more accurate and less ambiguous representation of the analyses.

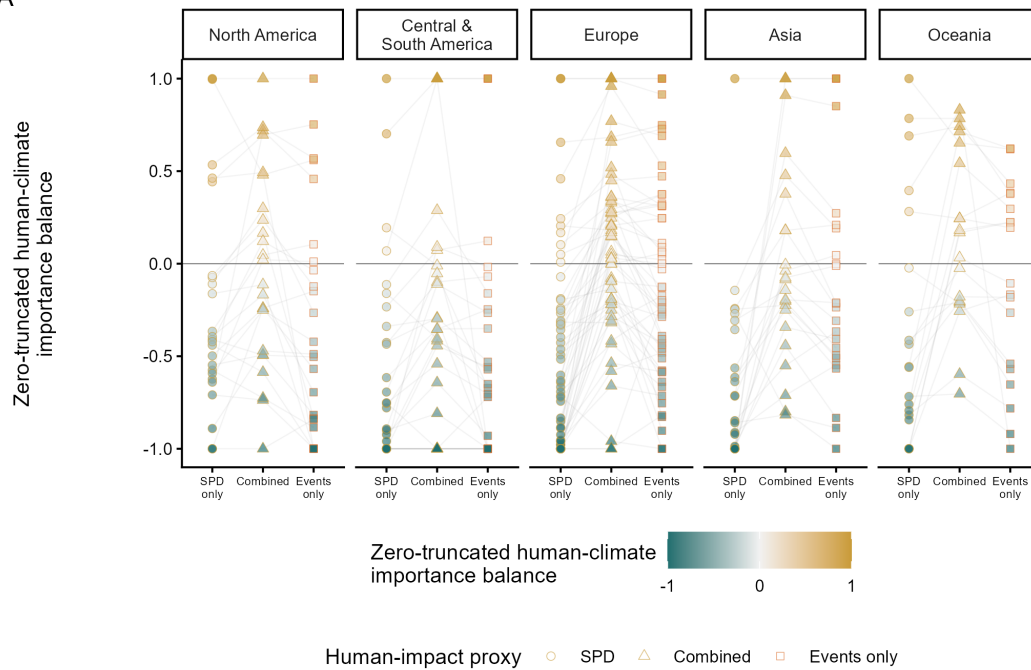
Response. We agree that the former title overstated both the temporal extent and the ecological domain of the study. The revised title is “Climate outweighs human effects on pollen assemblage dynamics from 8.5 to 2.0 ka BP,” and the principal conclusion is restricted to this interval. We have retained the expert event chronology as secondary evidence of human land use because it represents a different historical source from the archaeological radiocarbon SPD and includes indications of disturbance that the SPD cannot identify. Its categories are necessarily coarse and do not describe the full range of local practices. We therefore repeated the matched H1 comparisons with SPD alone, events alone, and both sources together. Across 151 spatial sequences and 61 continental-group-age comparisons, climate had the larger median estimated contribution with each representation of human activity (Supplementary Figs. S9 and S39). The combined spatial estimate was close to parity, and individual sequences and time intervals varied; the event result does not extend the principal SPD conclusion into younger periods. Because the events were interpreted in part from pollen diagrams, this comparison tests sensitivity to the human proxy rather than providing independent confirmation of the pollen response.

Changes in the revised article. The title, Abstract, hypotheses, Results, Discussion, and Conclusion now define the principal interval as 8.5–2.0 ka BP. Younger event observations are labelled separately in Fig. 4. The Methods and Supplementary Information explain the different historical evidence represented by SPDs and expert-interpreted pollen events.

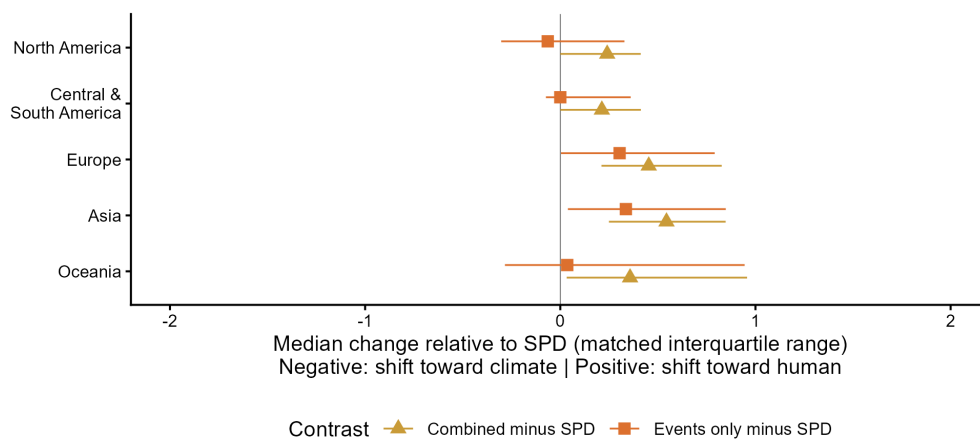
The matched comparisons are reproduced here to show the effect of representing human activity with archaeological dates, pollen-interpreted events, or both.

Regional-event sensitivity of spatial human-climate balance

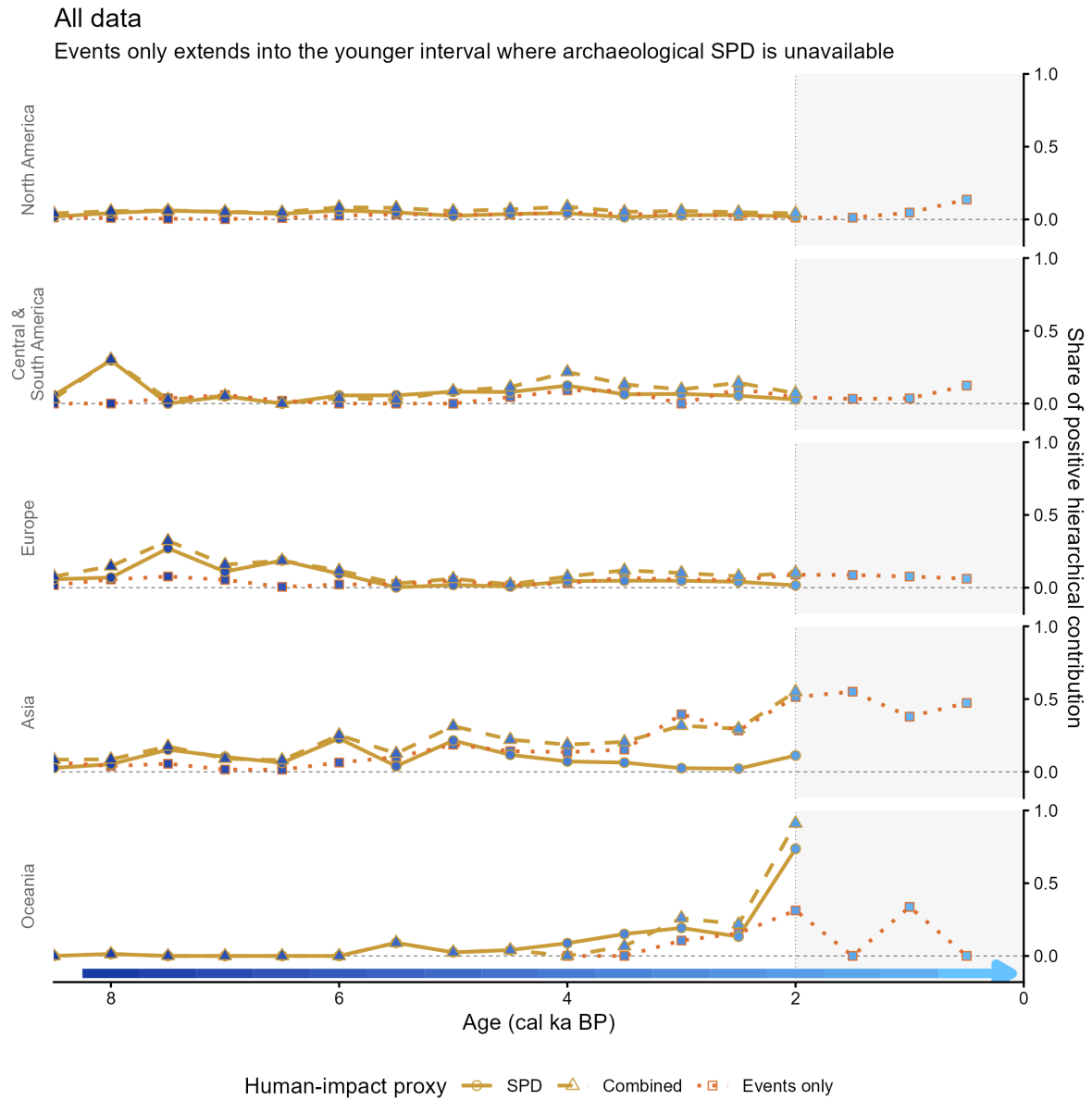
A



B



Supplementary Fig. S9 (reproduced). Spatial human-proxy sensitivity. The three representations are compared within the same sequences with observed event histories; event evidence is partly derived from pollen diagrams.



Supplementary Fig. S39 (reproduced). Temporal human-proxy sensitivity. The matched comparison uses continental-group time slices for which all three representations are available. Event-only profiles also extend into younger intervals beyond the principal SPD analysis.

Data presentation

Though two records are presented in Extended Fig. 3, a key figure missing from this work is a presentation of the full set of PAPs through time for each region. The authors have presented the set of predictor variables, rather than the response variables, which, in my view, are of substantially less importance than the response variables themselves. I find it difficult to make sense of and interpret statistical analyses for which at least some abstraction of the response variables (here, PAPs) are not presented in a paper and thus, for me, these need to be included.

Response. We agree that the histories of the pollen assemblage properties should be visible before their possible drivers are considered. We have therefore moved the descriptive trajectories ahead of the H1 results. Fig. 2 brings together six properties spanning richness, diversity, compositional change, and change-point density; Supplementary Figs. S12–S21 show all ten properties individually for each region within the continental groups. The shaded intervals convey uncertainty in the modelled trajectories, and gaps indicate where the contributing sequences do not provide sufficient temporal coverage. Together, these figures show both broad temporal tendencies and substantial variation among regions, providing ecological context for the subsequent comparison of climate and human activity.

The regional curves are not response data for either H1 analysis. In the spatial analysis, HVarPart is fitted to PAP values within each individual fossil pollen sequence, and the resulting contributions are summarised across sequences only afterward. In the temporal analysis, PAP values from individual sequences enter 500-year continental-group time slices, with spatial structure among those sequences represented in the model. Neither analysis uses the plotted regional summary curves as its response. The [full collection of figures for eligible individual sequences](#) is available on GitHub for inspection; the number of figures makes their inclusion in the Supplementary Information impractical.

Changes in the revised article. The Results now begin with “Temporal patterns in pollen assemblage properties.” Fig. 2 and Supplementary Figs. S12–S21 present descriptive regional trajectories. The Methods explicitly distinguish these displays from the sequence-level PAP values used in H1.

For direct assessment, we reproduce main Fig. 2 and the complete plots for two individual sequences also used to make Supplementary Fig. S11. Fig. 2 describes regional summaries; the individual plots show the underlying sequence-level histories available for the H1 analyses.

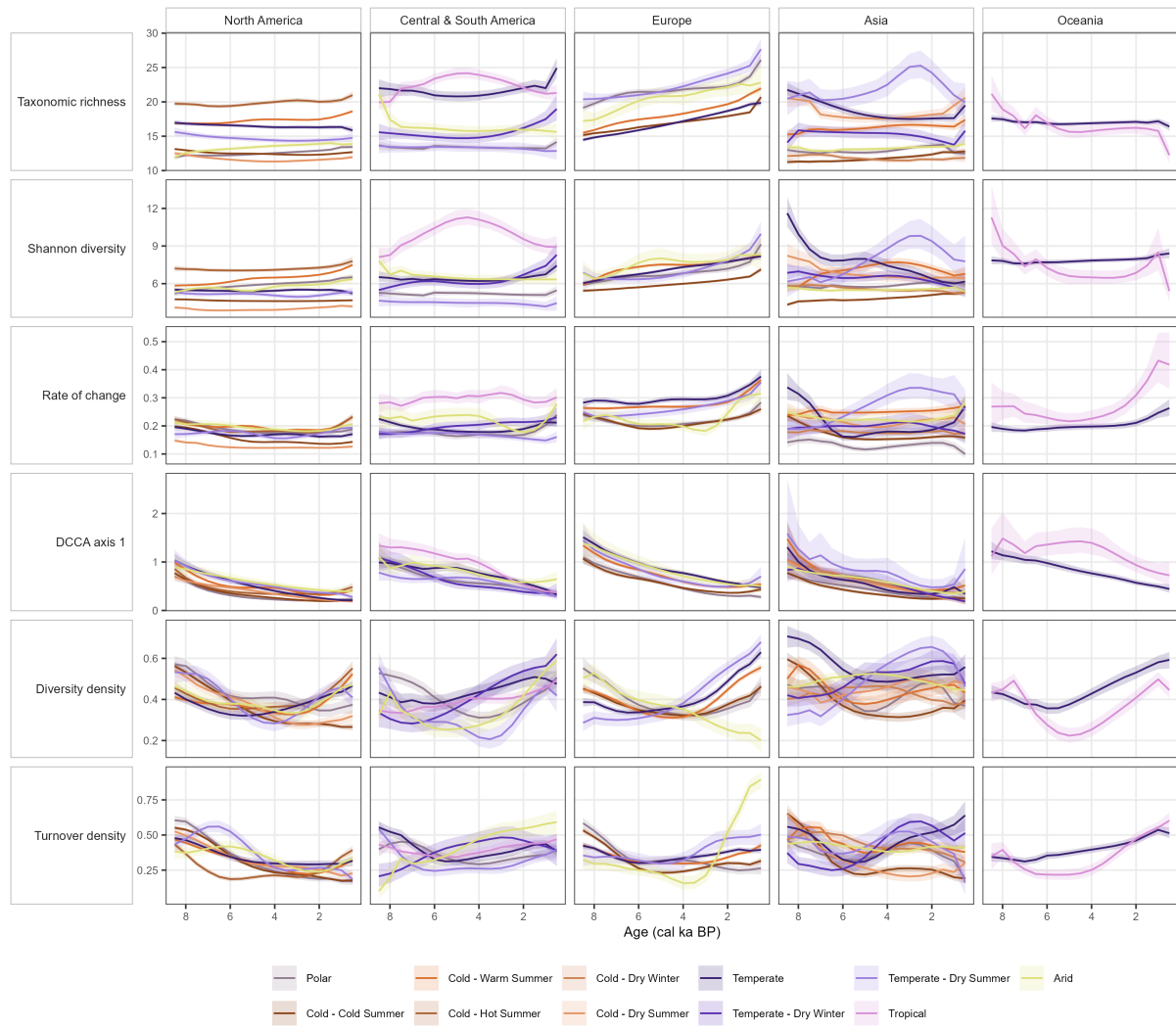
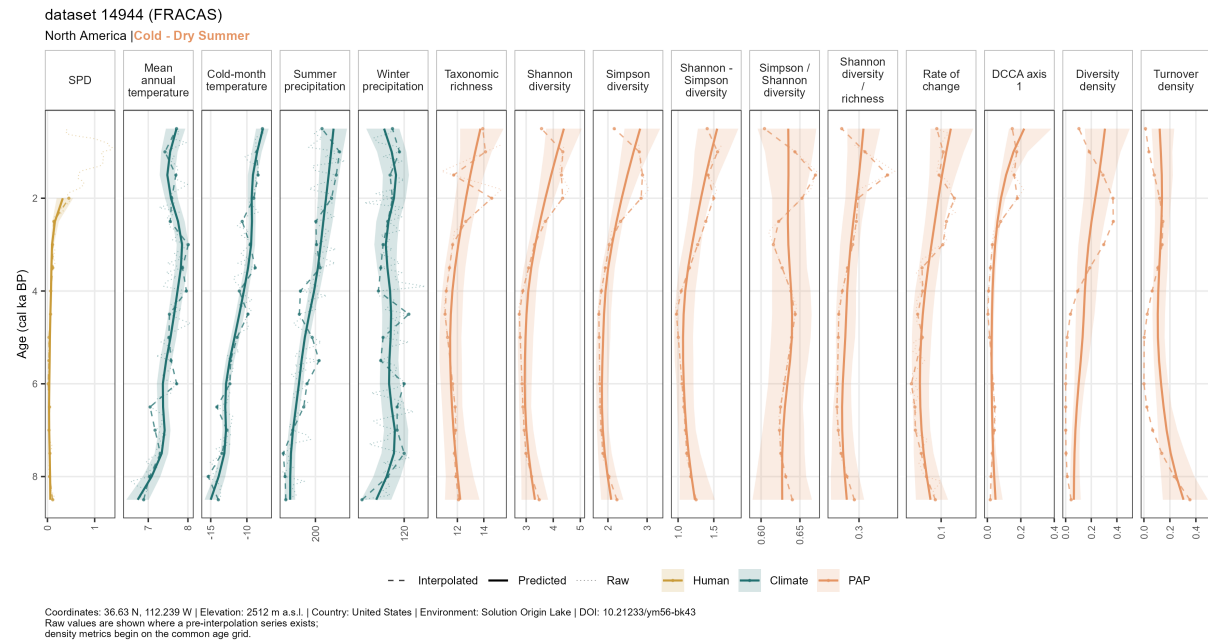
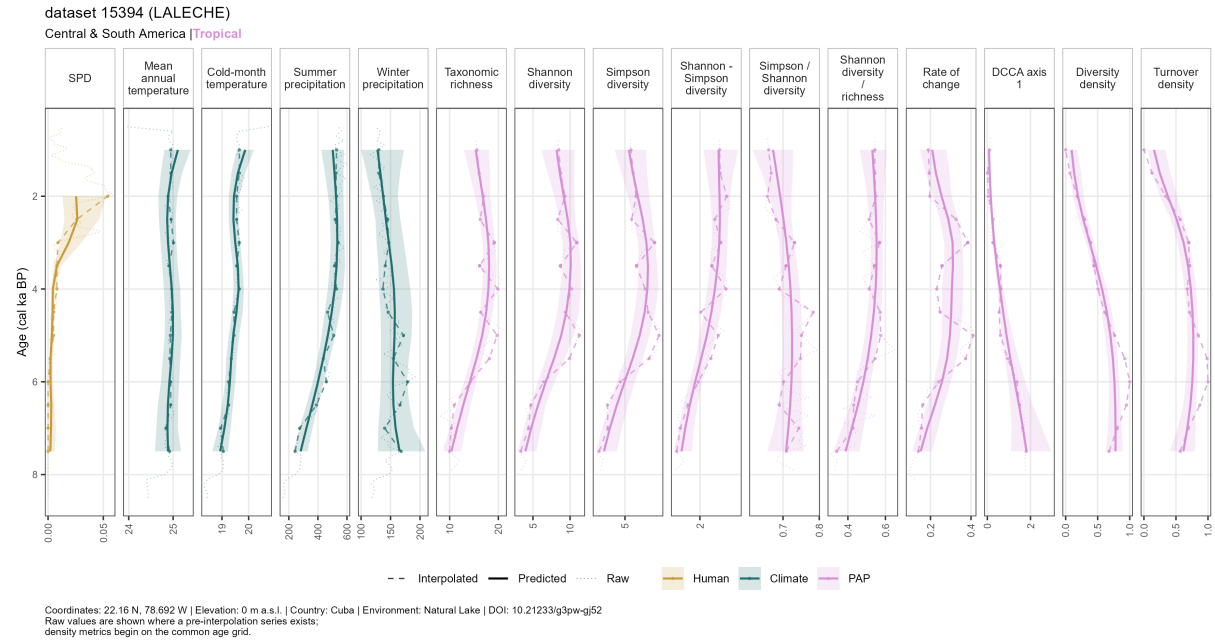


Fig. 2 (reproduced). Regional histories of pollen assemblage properties. These summaries provide ecological context and are not the response curves fitted in H1.



Individual sequence 14944 (FRACAS). Complete predictor and pollen assemblage property histories for one of the sequences illustrated in Supplementary Fig. S11. The full set of sequence figures is available at the [GitHub link](#) above.



Individual sequence 15394 (LALECHE). The second complete sequence plot used for Supplementary Fig. S11; it illustrates how the histories presented in the regional summary appear for an individual sequence.

Pollen diversity biases

Whether implicitly or explicitly stated, pollen data in studies such as these are used to make inferences on the changing diversity patterns, and relationships with drivers, of the parent vegetative assemblages of pollen samples (as stated by the authors, L652). Thus, the treatment of the pollen data in order to make robust inferences about the actual focus – the parent vegetative assemblage – is of paramount importance. For the reasons outlined below, the treatment of the pollen data in these analyses means that I find it difficult to have full confidence in the robustness of the computed PAPs and thus confidence in the conclusions of this work.

There is substantial variation in pollen productivity among pollen producing plants, which has the capacity to strongly bias estimates of individual PAPs, which the authors acknowledge (L687). Pollen grains from high pollen producing plants are more likely to be deposited and identified in pollen samples than pollen grains from low pollen producing plants. The result is that at low pollen sums, pollen assemblages (samples) are likely to be swamped by pollen grains from a relatively smaller number of high pollen producing plants. This means that the evenness of these samples will be low (high representation by a small number of types), the result of which is a reduced likelihood of detecting pollen types from other pollen producers, which thus artificially depresses the richness of these samples (e.g. Giesecke et al, 2014). Thus at low pollen sums, the richness of samples primarily reflects the evenness of those samples. Similarly, if the likelihood of sampling a diverse set of grains per sample is reduced (given the low evenness of samples), compositional turnover estimates are likely to be lower for these samples than between samples with higher evenness values. In these analyses, pollen samples are rarefied to either 150 or 25 pollen grains. Even 150 grains is substantially lower than the minimum rarefaction value of ~300-500 grains commonly used in pollen diversity analyses. For those samples rarefied to 25 grains, diversity estimates will almost certainly be affected strongly by these confounding issues. Extended Fig. 3 describes the PAPs and predictor variables for two records, where these described issues can clearly be seen: those Hill numbers N0, N1 and N2 (~richness, ~diversity and ~evenness) are strongly correlated.

Response. We agree that pollen productivity, dispersal, taxonomic resolution, basin setting, and count size all influence the ecological meaning of pollen assemblage properties. Low counts can make uncommon pollen types less likely to be detected and can affect estimates of richness, evenness, and compositional change. These properties are therefore interpreted as features of the recovered pollen assemblage rather than direct estimates of plant abundance or vegetation diversity. A minimum of 25 terrestrial grains determines whether an individual count-based sample can be retained; it is not a common rarefaction target imposed on every sample. Only 84 of 72,191 retained count-based samples (0.12%) contain 25–49 grains, whereas 98.7% contain at least 150 grains (Supplementary Fig. S23A). A count-based sequence is retained only when more than half of its samples in the study interval have at least 150 grains. The usual rarefaction target is 150 grains, with a lower sequence-specific target where a retained sequence cannot consistently support 150. This

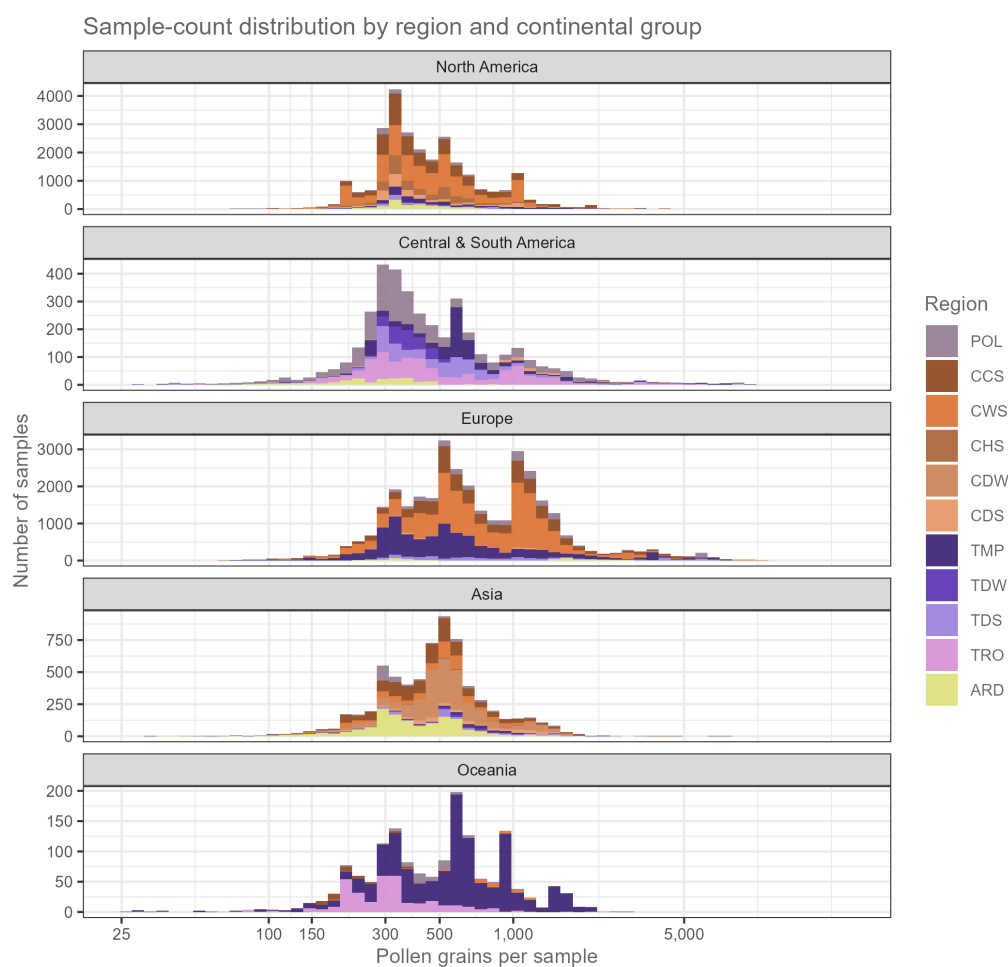
rule limits the contribution of low-count samples while preserving sequences from less well represented regions; it does not remove the uncertainty associated with smaller counts. We examined what would be lost if more demanding count criteria were applied to the 1,270 prepared sequences. A 300-grain rarefaction target, paired with a 50-grain sample minimum, retains 1105 sequences. A 500-grain target, paired with a sample minimum of about 83 grains, retains 616; a 1,000-grain target, paired with a minimum of about 167 grains, retains only 246. The 500-grain scenario therefore removes 51.5% of baseline sequences and 45.3% of their samples; the 1,000-grain scenario removes 80.6% of sequences. The losses differ among continental groups, so higher count requirements would come at the cost of a narrower and less geographically representative historical archive. Supplementary Fig. S23A shows the observed count distribution in count-based samples, while Fig. S23B shows that the loss of sequences and samples differs markedly among continental groups. We retain the baseline rule on that basis while acknowledging the greater uncertainty of PAP estimates from smaller counts. The reviewer also notes that N_0 , N_1 , and N_2 covary in the example sequences. These indices describe related features of the same pollen assemblages, so their association alone does not establish that low counts caused the pattern. The count distribution and retention comparison above address the sampling concern directly.

These concerns can be at least partly mitigated against by performing some transformation of pollen counts based on relative pollen production and dispersal (etc). There are a number of methods to achieve this, the most notable of which is the REVEALS model, for which there are the requisite datasets for many of the regions used here. However, this is not implemented in these analyses. At the very minimum, I would expect to see a supplementary analysis that considers only the subset of samples and records with a substantially higher minimum pollen grain N (300-500 grains), to check that results are not driven by those issues relating to relative differences in pollen production and dispersal described above. I am of the view that all samples with <150 grains should be excluded from these analyses.

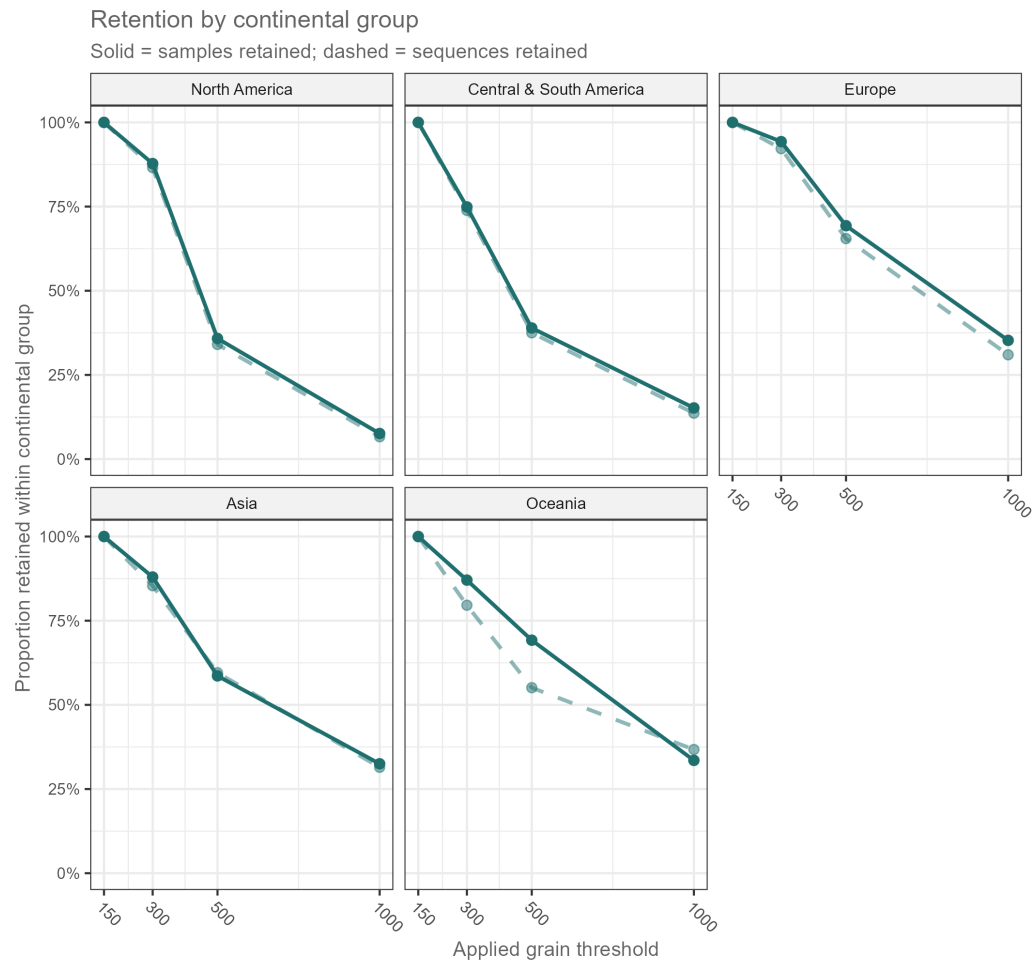
Response. REVEALS can improve estimates of vegetation composition where the necessary information is available, but it cannot currently provide a comparable correction for this global compilation. It requires taxon-specific pollen-productivity estimates, choices about dispersal and relevant source area, and information on basin size and setting; their availability and transferability vary among regions and through time. Taxa without suitable productivity estimates would also be omitted. Applying REVEALS only where these inputs are well established would change both taxon coverage and the meaning of the response among continental groups. We therefore retain pollen assemblage properties as pollen-domain measures and discuss productivity and dispersal as limitations; rarefaction does not correct either bias.

Changes in the revised article. The Methods now explain the count procedure. Supplementary Fig. S23A shows the observed pollen counts, and Fig. S23B shows the loss of

geographical and temporal coverage under stricter targets. The Discussion addresses how pollen production, dispersal, and finite counts constrain ecological interpretation.



Supplementary Fig. S23A (reproduced). Terrestrial pollen counts in individual count-based samples, grouped by continental group and region. Percentage-only sequences are excluded because their values are not grain counts.



Supplementary Fig. S23B (reproduced). Proportions of sequences (dashed) and samples (solid) retained within each continental group. The 300-, 500-, and 1,000-grain labels denote rarefaction targets with associated sample minima of 50, about 83, and about 167 grains, relative to the 150-grain target and 25-grain sample-minimum baseline. This is a coverage diagnostic, not an H1 model refit.

SPD biases

SPDs of archaeologically attested radiocarbon dates are becoming increasingly used to infer relative changes in human paleodemography. However, there is substantial and ongoing debate in the literature among archaeologists about whether this method can actually tell us anything about paleodemography (e.g. Chaput and Gajewski, 2016; Crema and Bevan, 2020; Crema, 2022). It is only material from times and places of archaeological interest that are radiocarbon dated. Thus the number of radiocarbon dated objects through time and space is intensely non-random, clustering around locations and periods of interest to the archaeological community. This is likely the cause of the substantial apparent paleodemographic spikes during transitions between archaeological periods (archaeologists are interested in what caused and was affected by these transitions). On top of this, radiocarbon dating is expensive. So, when possible, other objects – e.g., pottery, coins – are preferentially used to date archaeological remains, which results in a reduction in the number of radiocarbon dates when these objects are readily available. This is most obviously shown in Europe from ~3 cal k BP, from which point the radiocarbon SPD shows a monotonic decline (Extended Data Fig. 4). This apparent reduction is entirely artificial and driven by the general cessation of the use of radiocarbon dating to date archaeological remains, given the proliferation of other dateable material across the continent from this period onwards. Given that this variable underpins this paper’s primary analyses and main conclusions, I would expect some critical discussion in the main text that outlines the well-known and strongly debated limitations of using SPDs for paleodemographic inference, though this is missing. This would then help justify the decision to cut off the analyses at 2.5 cal k BP, which at the moment does not feel sufficiently justified in the text.

Response. We retain archaeological SPDs as validated proxies for regional human population changes, while agreeing that they are not direct population censuses. They are shaped by the survival and discovery of archaeological material, the history and intensity of research, laboratory dating practices, and the structure of the radiocarbon calibration curve. To examine whether these limitations drive our conclusion, we repeated the matched H1 analyses with SPD alone, expert-interpreted human events alone, and both together. Climate had the larger median estimated contribution under all three definitions in 151 matched spatial sequences and 61 matched continental-group–age comparisons (Supplementary Figs. S9 and S39, reproduced above). Events provide evidence of land-use indicators distinct from the archaeological dating record, but they are partly derived from the pollen diagrams used to calculate PAPs. Agreement across proxies therefore supports the broad result against dependence on SPD alone; it is not an independent validation of the pollen response or proof that SPDs are free of bias. The regional distributions of dates and SPDs showed that 2.0 ka BP is the youngest common 500-year boundary at which a broad comparison remains supportable. We use it as a practical limit for this global historical comparison, while presenting younger event observations separately.

Changes in the revised article. The Methods and Discussion now describe the sampling limitations of SPDs and the distinct limitations of the event chronologies. Supplementary Figs. S9 and S39 show the matched human-proxy comparison, while Fig. S2 shows the regional coverage that informed the 2.0 ka BP boundary. The principal conclusion does not extend into younger periods.

Notwithstanding these comments, of most concern to me is the different spatial scales of the respective variables used in these analyses. The relevant source area of pollen for any given extracted fossil pollen sequence varies from meters up to tens of kilometers (rarely hundreds of km), depending on the deposition environment, location, etc. However, the authors pull radiocarbon dates from either 250km or 500km surrounding a pollen record extraction location to build their SPDs. These scales are vastly different and it is perhaps not surprising that PAPs from, for example, a bog site with a pollen source area in the order of hundreds of meters are not associated with SPDs built from radiocarbon dates pulled from within 250 or 500km but that they are associated with paleoclimate estimates for the actual location of a pollen record. In principle, either (i) a dynamic radius for retrieving radiocarbon dates per pollen record could be used (based on each record's pollen source area), (ii) these analyses could be run regionally (using averages across PAPs and predictor variables) or (iii), at the very least, a sensitivity analysis should be run that restricts the analyses to those records with the largest pollen source areas, with SPDs built using a radius for radiocarbon dates that is at least equal in order of magnitude to it.

Response. We agree that pollen and archaeological evidence describe landscapes at different scales. The 250 km archaeological neighbourhood is a pragmatic regional context, not an estimate of the pollen source area. Human groups, settlements, and areas of land use could shift during the centuries represented by a 500-year interval, movements that our time slices cannot resolve. Restricting archaeological dates to the immediate pollen source area could therefore miss relevant regional activity and, in many places, leave too few dates for a usable SPD. We chose 250 km to balance proximity and date coverage, with a 500 km fallback where the smaller neighbourhood was insufficient. We cannot tailor the radius to each pollen sequence because consistent information on basin size, depositional setting, and pollen source area is unavailable across the compilation.

A wider radius can also bring in evidence of people whose activities did not affect the vegetation surrounding a pollen site. It might thus overstate the relevance of regional human presence to that sequence, although its effect on the estimated human contribution cannot be assumed: the model depends on how the archaeological and pollen histories covary. We tested this directly by comparing strict 250 and 500 km SPDs for the same eligible data. Among 999 matched sequences, the median 500-minus-250 km change in the spatial human-minus-climate balance was 0.025, a modest shift toward the human proxy; 94 sequences reversed their climate-versus-human ranking. Median human allocation also

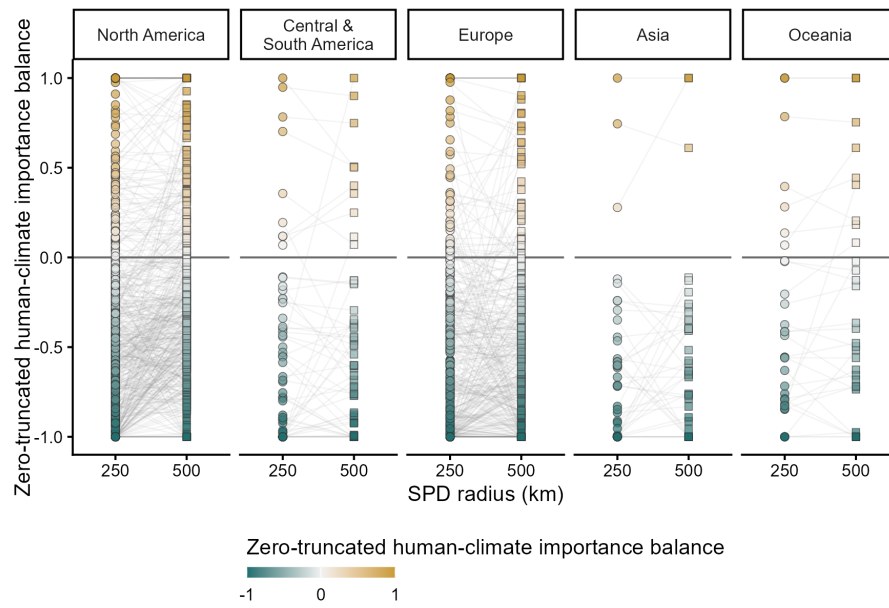
rose by 0.035 across 70 matched continental-group–age comparisons, of which 3 reversed ranking. Despite these shifts, climate retained the larger estimated contribution for most matched sequences and time slices (Supplementary Figs. S8 and S37–S38). Neither radius resolves the site-to-region scale mismatch, which limits inference about individual land-use histories.

Changes in the revised article. The Methods now explain the regional and coverage rationale for the 250 km neighbourhood and the 500 km fallback. Supplementary Figs. S8 and S37–S38 and Supplementary Table S9 present the matched-radius comparison. The Discussion explains why that comparison cannot substitute for site-specific pollen-source estimates.

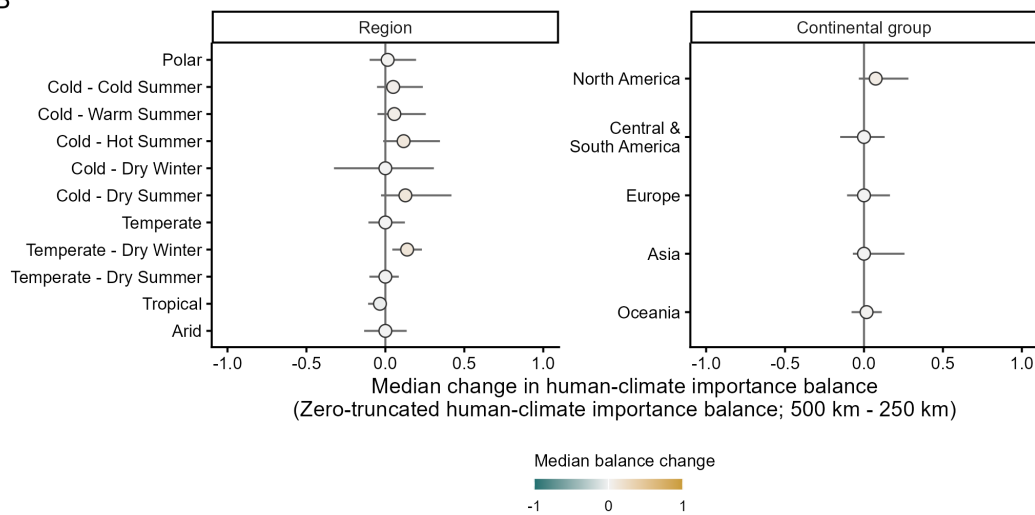
The matched spatial and temporal comparisons of archaeological search radii are reproduced below. They test the operational radius choice; neither radius represents a measured pollen-source area.

Sensitivity of spatial human-climate importance balance to SPD search radius

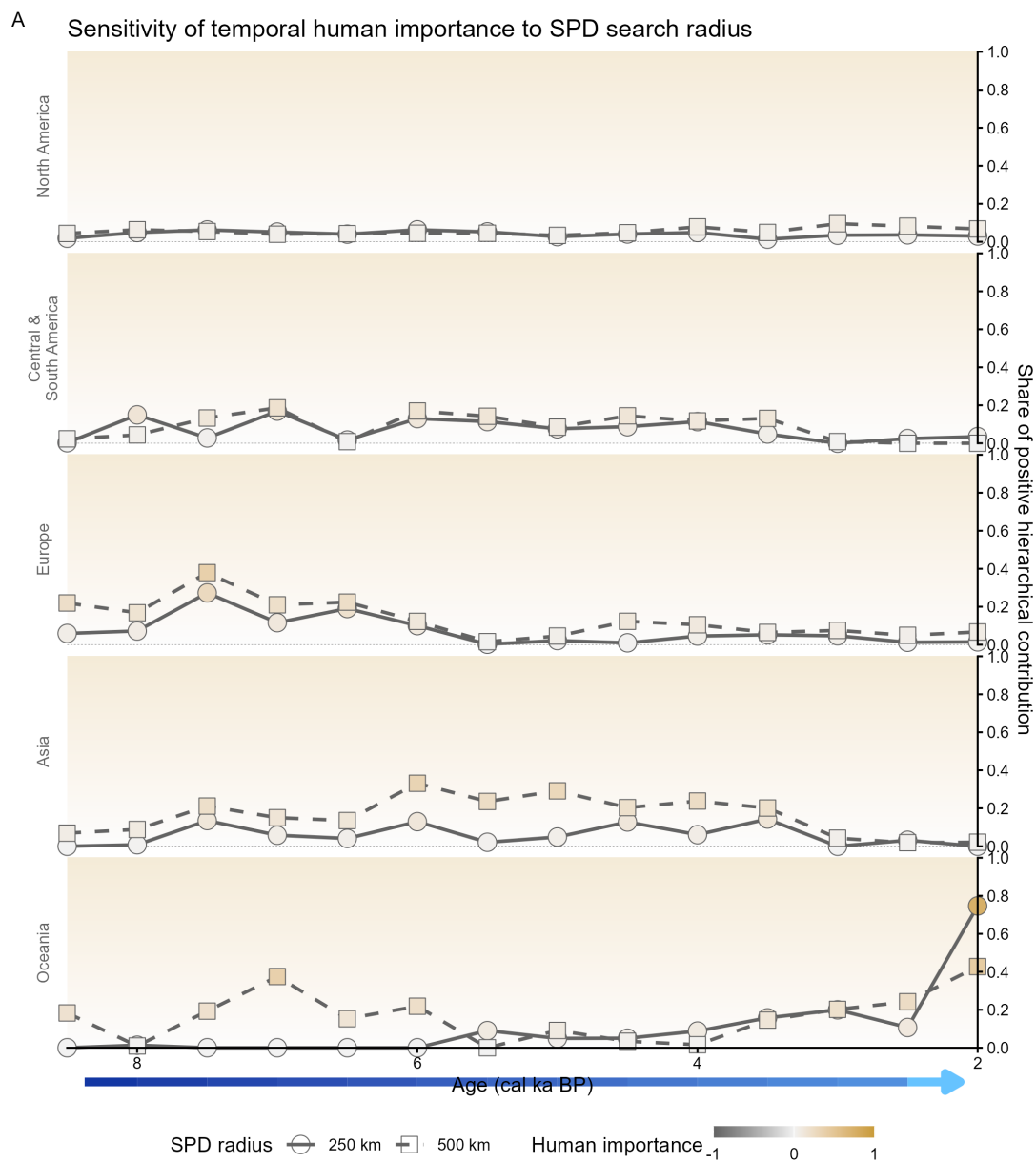
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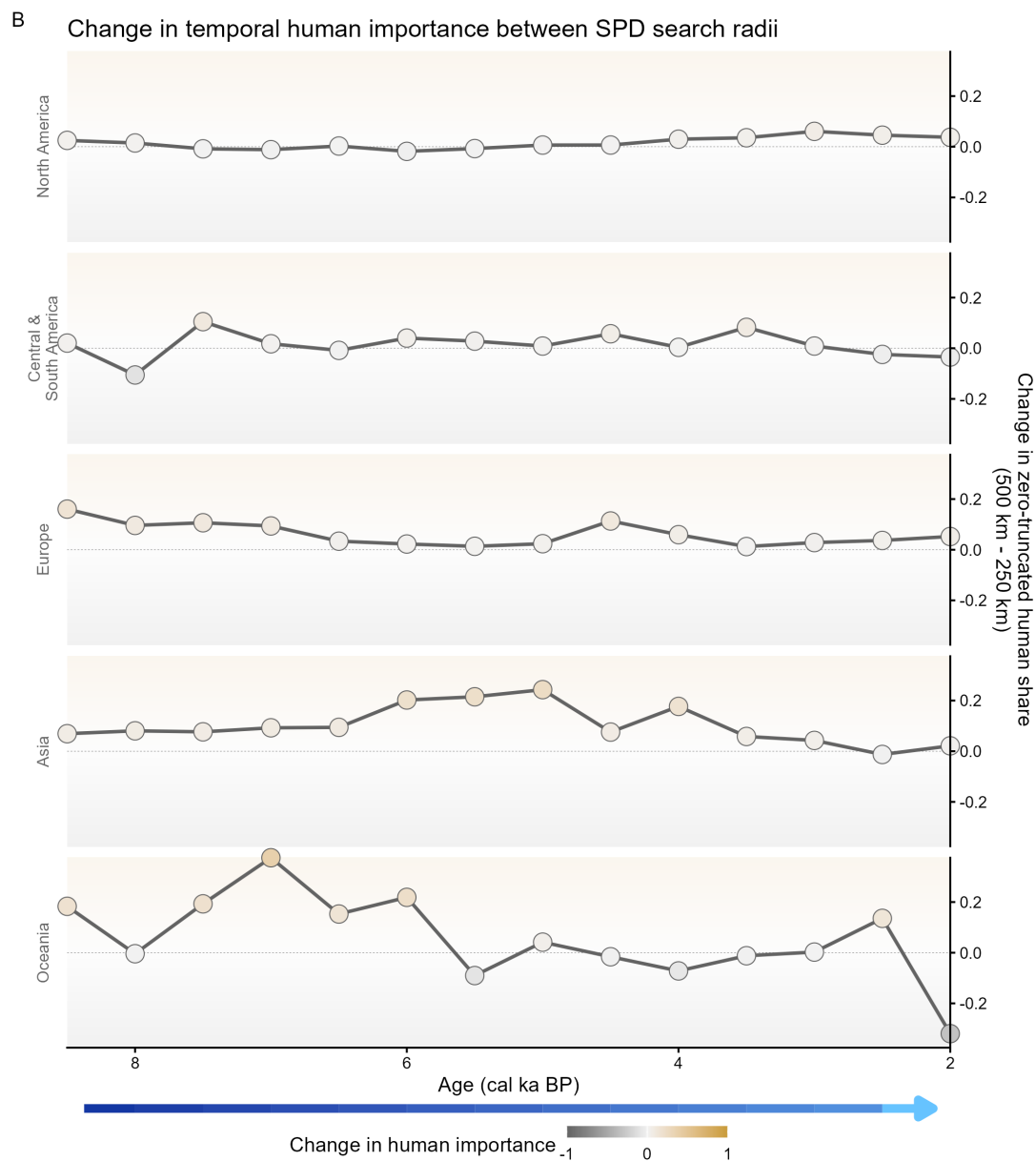
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Supplementary Fig. S8 (reproduced). Spatial SPD-radius sensitivity. The same eligible pollen sequences are compared using strict 250 and 500 km archaeological neighbourhoods; a negative balance indicates the larger estimated climatic contribution.



Supplementary Fig. S37 (reproduced). Temporal SPD-radius profiles. Spatially controlled H1 allocations are shown for matched continental-group-age comparisons under strict 250 and 500 km archaeological neighbourhoods.



Supplementary Fig. S38 (reproduced). Temporal change with SPD radius. Values are the 500-minus-250 km change in the estimated human allocation for matched continental-group-age comparisons; positive values indicate a shift toward the human proxy.

Human events variable

Even by itself, the development of this variable represents a major undertaking and whilst I appreciate the substantial effort, I find the decomposition of continuous human change into categorical events represented by a binary variable overly simplistic and almost certain not to capture diverse changes across the PAPs considered here. I also find there to be a lack of description and justification of the method. Which experts were asked? Why were these experts asked? Who do they represent? Why and how were these events chosen? By whom? What were the exact criteria used for each event * region considered? Indeed, for Central and South America, the authors use an established index from the literature, which underlines the extent of the work required for the development of indices such as these for individual continents, let alone for all continents as is attempted here.

The reductive nature of the index is exemplified best for North America, for which the authors provide three event categories: No Evidence of Impact, First Cultivation and European Settlement. For many Indigenous North American peoples, cultivating domesticated plant types remained supplementary to a much broader focus on foraging wild foods (both plant and animal) over the Holocene. Wild foods were often foraged with the substantial assistance of fire, which was used extensively across the continent before, during and after cultivation of domestic crops. According to this index, there was no human impact in North America before First cultivation, which is certainly untrue. Indigenous groups across North America have impacted vegetative communities over the entire Holocene and into the Pleistocene both directly, through burning, and indirectly, through the removal of the largest animals. Both of these would have strongly impacted vegetative communities, but their effects are entirely missed by the coarse categories here. It would be a vast amount of work to develop a pollen-based index of human impact for all regions globally that was transparent and not overly reductive. I do not believe that, in its present form, the inclusion of this binary variable adds to this work and I would simply remove it and curtail the analyses at the end of the period of SPD relevance, however defined.

Response. We agree that a binary event chronology cannot capture the continuity and diversity of past human land use. Clearance, grazing, cultivation, burning, and settlement varied in form, duration, intensity, and ecological consequence among regions. We have therefore removed “No Evidence of Impact” as an interpretation: absence of a recorded event means only that the specified pollen indicator was not identified. Nevertheless, we retain the chronology as secondary historical evidence because it brings locally interpreted indications of land use into a study otherwise dependent on broad archaeological SPDs, includes some younger observations, and allows us to ask whether the climate–human comparison changes when human activity is represented differently. Its partial dependence on the pollen sequence is stated explicitly, and it is not presented as independent confirmation.

The comparison is limited to places and periods for which an expert chronology was actually available: 151 pollen sequences in the spatial analysis and 61 continental-group-age combinations in the temporal analysis. Adding or substituting the event evidence changed estimates in both directions. This variability is historically informative because it shows that the inferred human contribution depends in part on the archive used to represent past activity. It does not extend the principal SPD conclusion beyond 2.0 ka BP. John Birks and Hilary Birks assessed Europe and North America; John Birks and Kuber Bhatta assessed Asia; and Simon Haberle and Annika Herbert assessed Oceania. They were chosen for regional palaeoecological and pollen-analytical expertise. The Central and South American indicators follow the published regional synthesis. The Methods section describes how the diagrams were assessed, and Supplementary Table S4 summarises the event categories, and the Methods retain the detailed regional detection criteria from the submitted version.

Changes in the revised article. The event chronology is now presented as a complementary historical perspective rather than evidence equal to the archaeological SPD. The Methods identify the experts and regional indicators, the Discussion addresses the incompleteness and partial circularity of the evidence, and the Supplementary Information compares results based on SPDs, events, and both sources together.

The matched SPD, event, and combined comparison is reproduced with our response to the reviewer's opening concern above.

Statistical analyses

Firstly, I cannot see a summary of the total variance explained by the models here. This is of crucial importance because if the models explain a very small amount of the total variation in the data, then it would be difficult to say that humans/climate are most/not important drivers of the PAPs, in my view. These should be added as a Supplementary table and summarised in some way - perhaps across records, by region - as an Extended Data Figure.

Response. We agree that the relative importance of climate and human activity must be interpreted alongside the amount of pollen assemblage variation explained by the complete model. The revised Results and Supplementary Table S5 now report total adjusted R^2 , while Supplementary Fig. S31 shows its distribution among individual sequences, continental groups, and regions. Supplementary Fig. S32 shows how model fit relates to the estimated human contribution. A larger climatic allocation within a fitted model does not mean that climate explains most of all observed assemblage change, particularly for sequences with limited model fit.

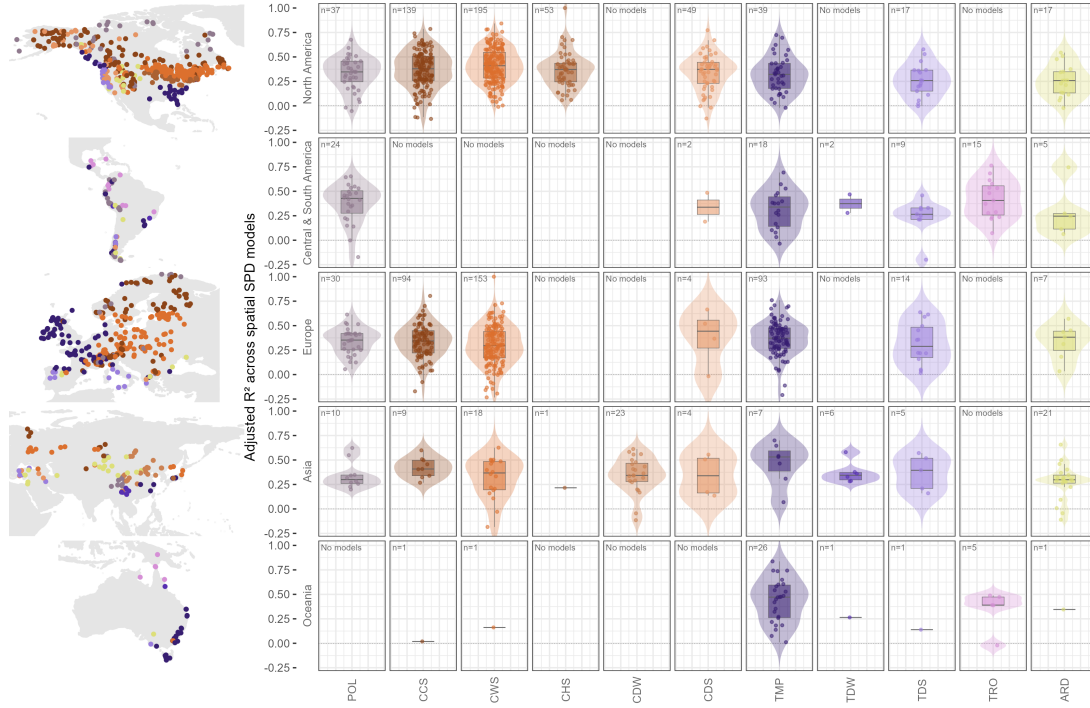
We also corrected the extraction of predictor contributions from the complete hierarchical variation partition. The **signed profile** retains positive and negative contributions from

each model. Negative components can arise when climate, human activity, and the control variables share information or suppress one another in the statistical decomposition; they do not mean that a driver explains a negative amount of ecological change. The **zero-truncated profile** sets negative individual contributions to zero and rescales the remaining components to a non-negative composition. We use this profile in the main figures because it makes the relative climate and human allocations easier to compare visually across sequences and time slices. Its bars describe a display composition, not the fraction of total pollen variation explained, and a zero bar does not establish that the driver had no effect. The **exclude-negative profile** instead omits an entire model whenever its decomposition contains a negative individual contribution, then summarises the remaining eligible models. It avoids truncating those contributions but changes the subset of sequences or time slices being compared. Supplementary Fig. S4 shows the three profiles side by side.

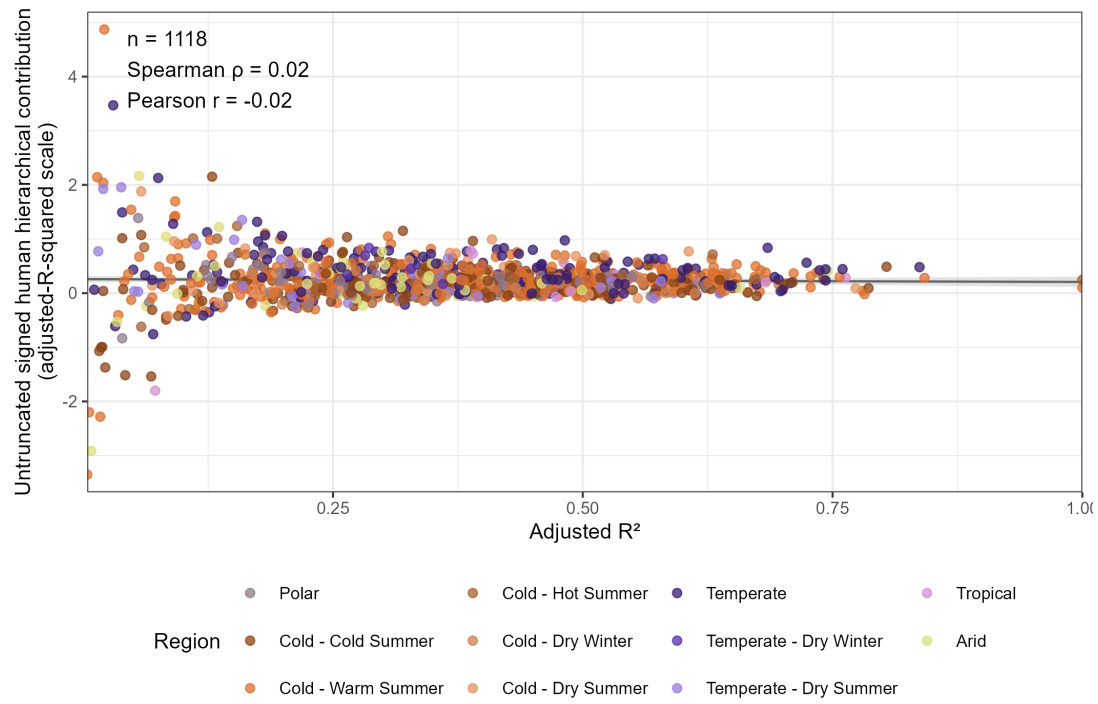
We repeated the principal spatial H1, temporal H1, and H2 attribution displays using signed contributions (Supplementary Figs. S5, S22, and S33). The broad result that climate had the larger estimated contribution remains under the signed presentation, although some individual sequences, time slices, and regions differ. The signed figures retain the negative components concealed by the zero-truncated main displays and allow the reader to assess those differences directly.

Changes in the revised article. The Results now present total model fit alongside relative predictor importance. The Methods distinguish total adjusted R^2 from hierarchical contributions and explain why the zero-truncated composition is used for the main visualisations. Supplementary Figs. S4, S5, S22, S31–S33 provide the extraction-profile comparison, signed counterparts, and model-fit diagnostics.

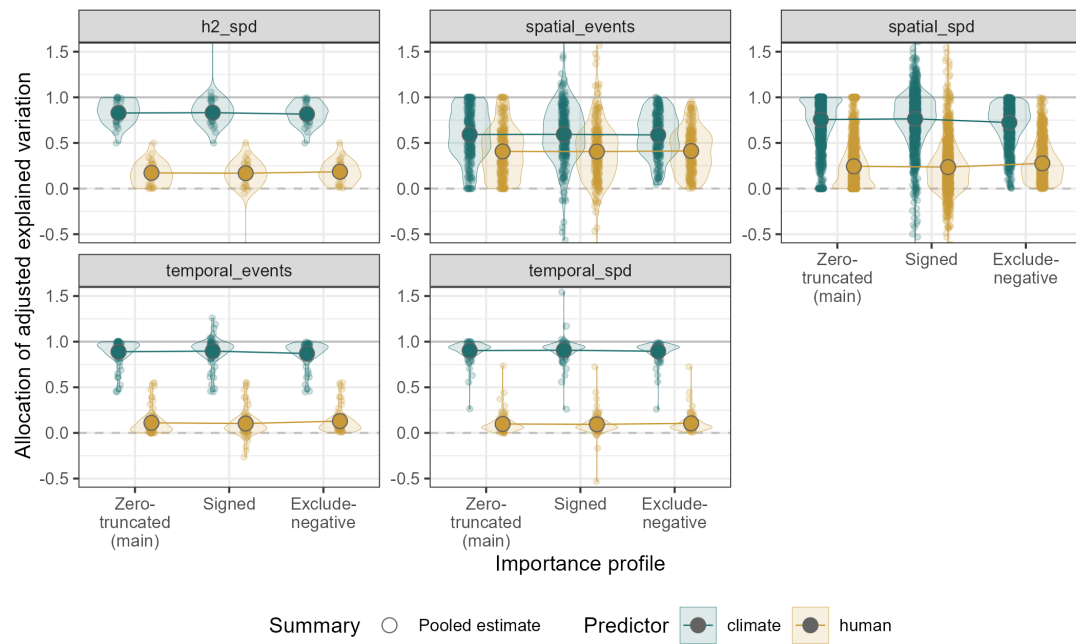
We reproduce the model-fit and extraction-profile figures, followed by the signed versions of the principal attribution displays.



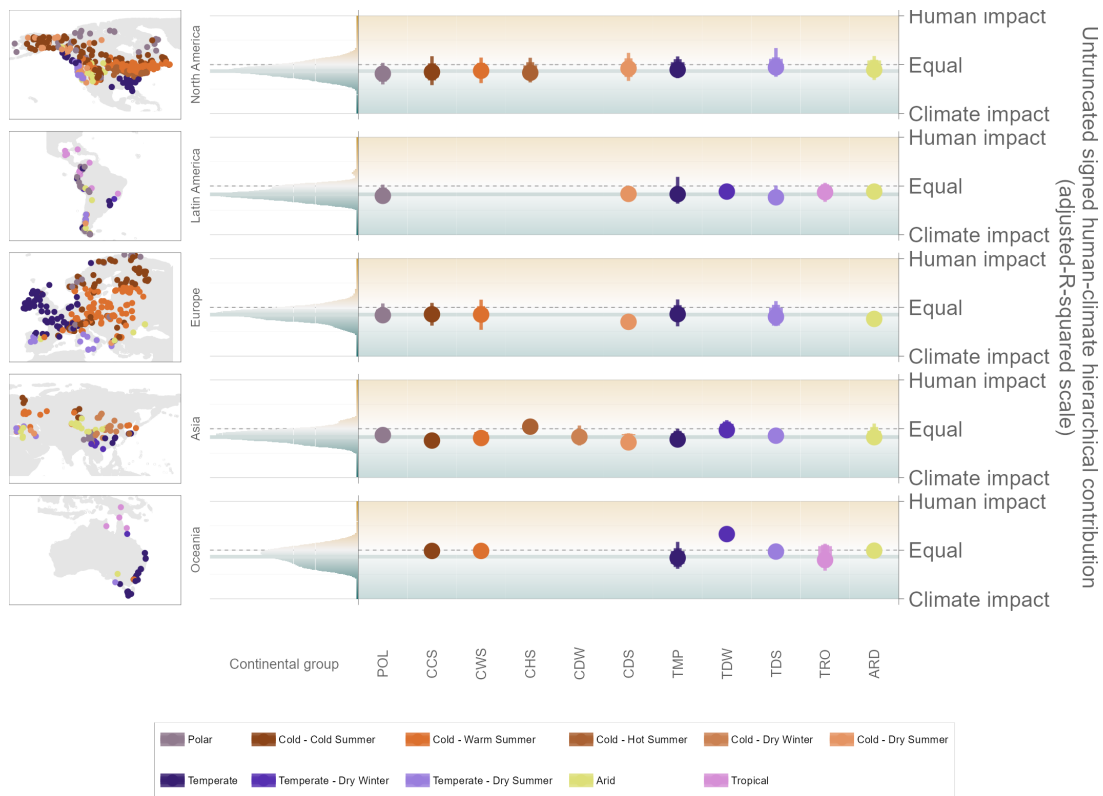
Supplementary Fig. S31 (reproduced). Total adjusted R^2 among pollen sequences. The distribution shows variation in complete-model fit across continental groups and regions; negative adjusted values remain visible.



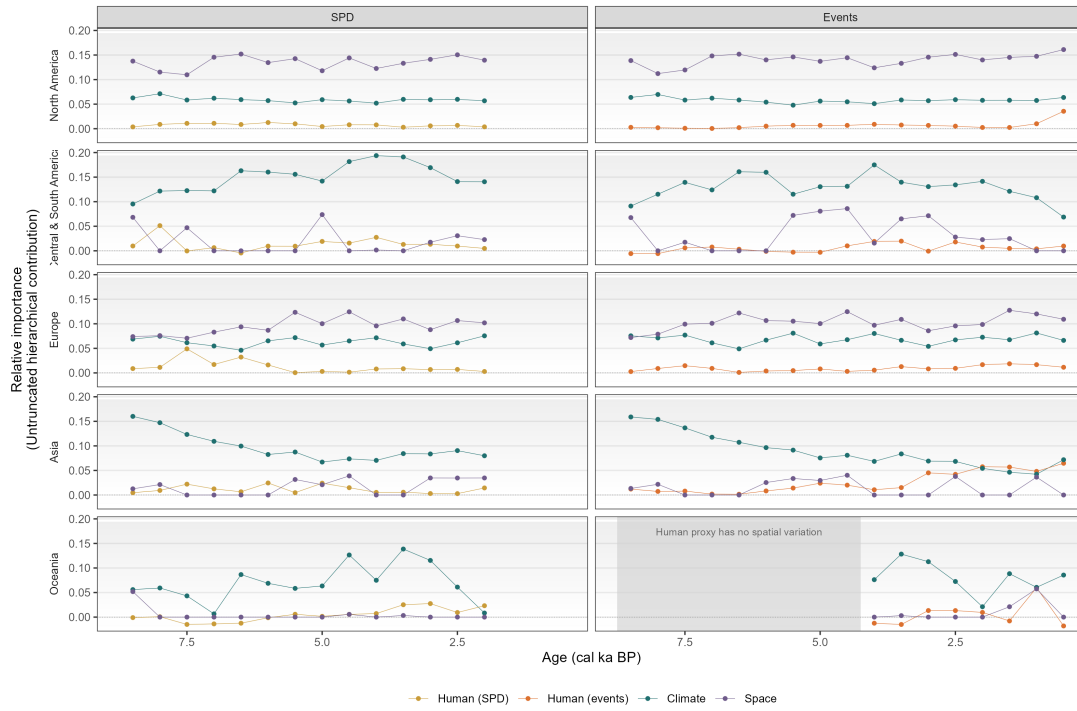
Supplementary Fig. S32 (reproduced). Model fit and human contribution. Total adjusted R^2 is compared with the signed human hierarchical contribution for eligible spatial SPD models; these are distinct model summaries.



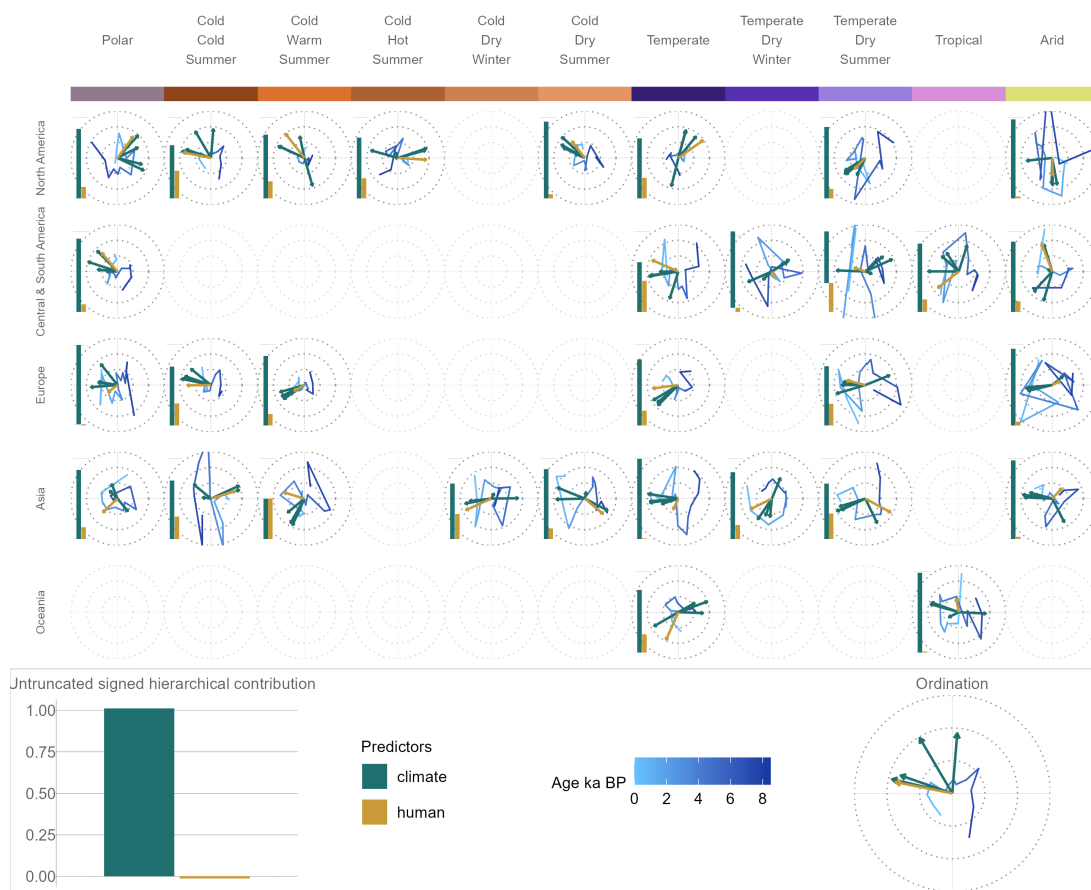
Supplementary Fig. S4 (reproduced). HVarPart extraction profiles. Signed, zero-truncated, and exclude-negative summaries are compared for their respective eligible analysis subsets; these allocations are distinct from total adjusted R^2 .



Supplementary Fig. S5 (reproduced). Signed spatial H1 result. The human-minus-climate balance uses the same sequences and time control as main Fig. 3. Negative signed components retain information hidden in the zero-truncated display.



Supplementary Fig. S22 (reproduced). Signed temporal H1 result. The continental-group time slices and spatial control match main Fig. 4; missing comparisons remain non-estimable.



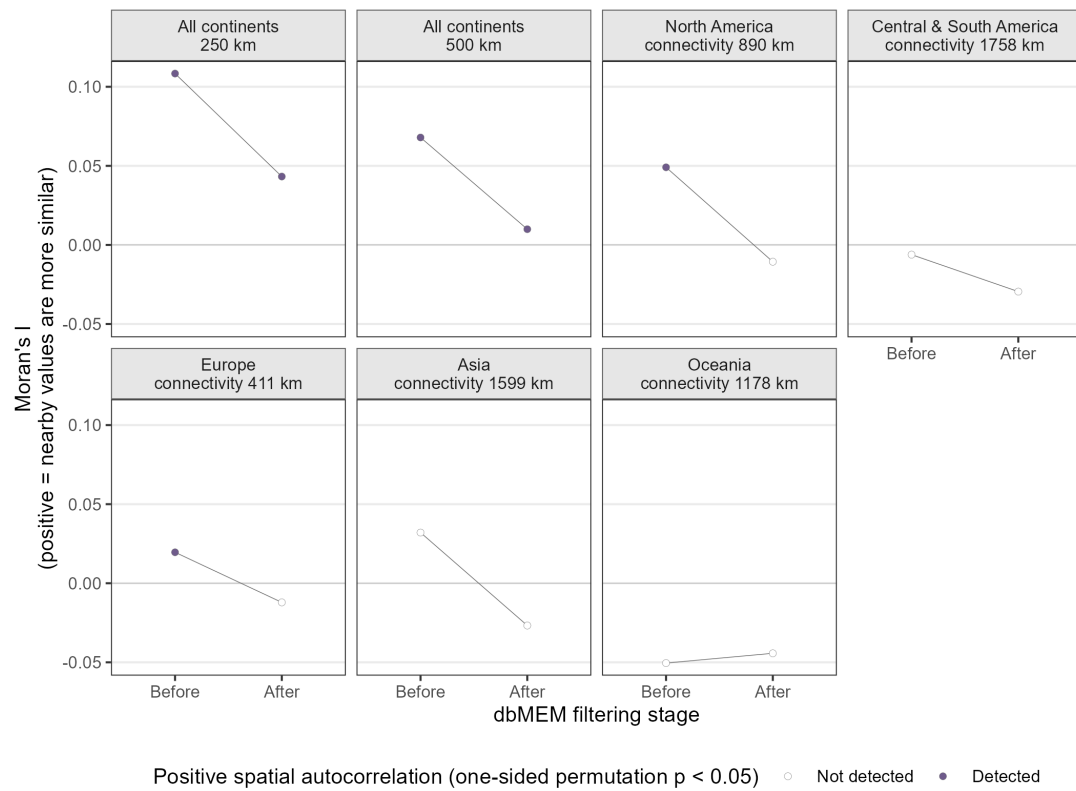
Supplementary Fig. S33 (reproduced). Signed H2 result. The analysis groups match Supplementary Fig. S10. Signed contributions retain negative components rather than setting them to zero for display.

Secondly, temporal and spatial autocorrelation do not seem to be accounted for in these models, and their effects not checked for. These effects potentially confound model inferences and thus either an explanation as to why they were not included needs to be added to the methods, or some additional variables need to be added to the models. This is likely of most importance to those analyses in Fig 2 where there will undoubtedly be a substantial trend component to the PAPs over the long period considered.

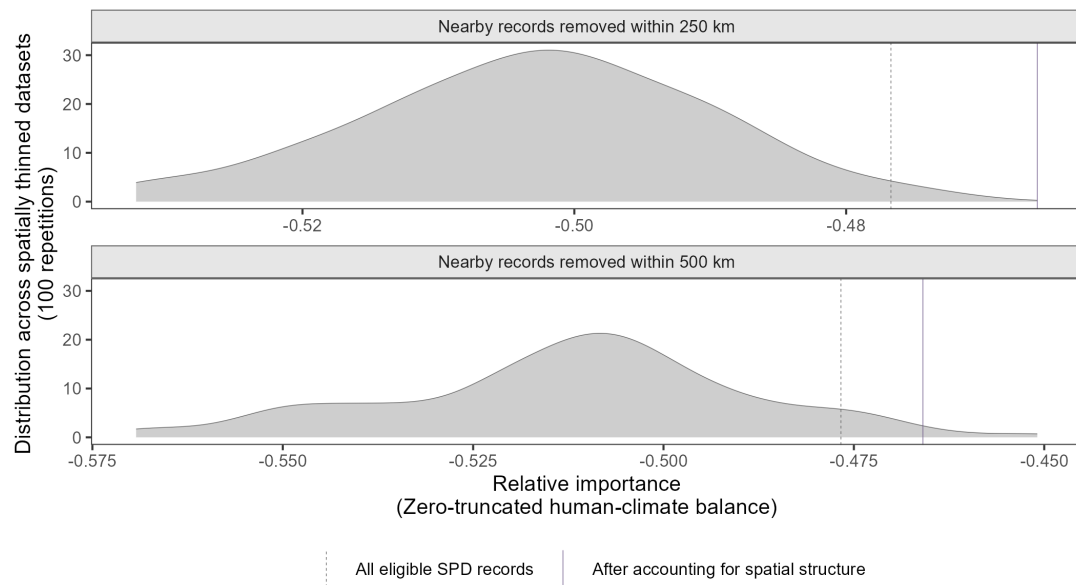
Response. We agree that ecological observations close together in time or space are not independent. Long-term succession and directional change can make neighbouring levels within a pollen sequence resemble one another, while shared geography can produce similar assemblages among nearby sites. The principal analysis of individual sequences now accounts for temporal structure, and the continental-group analysis through time accounts for spatial structure among sites. We also examine remaining spatial pattern in the residuals, repeat analyses using more widely spaced sequences, and test whether the broad result depends on any single continental group or region. These checks retain the general predominance of climate, while showing that some individual estimates remain uncertain or sensitive to analytical choices.

Changes in the revised article. Figs. 3 and 4 now present the analyses that account for temporal or spatial structure. The Methods describe these controls, and the Supplementary Information reports checks for remaining geographical pattern, the influence of closely spaced sites, and the effect of excluding individual continental groups or regions.

We reproduce the residual spatial-dependence and thinning diagnostics because they show what the controls changed and what geographical structure remains.



Supplementary Fig. S6 (reproduced). Residual spatial dependence. Moran diagnostics compare spatial pattern before and after control in eligible spatial subsets; residual clustering is not assumed to disappear completely.



Each value compares human and climate influence in one thinned SPD dataset. Negative values indicate greater climate influence.

Supplementary Fig. S34 (reproduced). Spatial thinning. Repeatedly retaining more widely separated sequences tests whether densely sampled places determine the aggregate climate–human balance.

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Response. We thank the reviewer for these references. The revised article uses Giesecke et al. (2019), Gordon et al. (2024), and Mottl et al. (2021) to place the results within later-Holocene histories of pollen diversity, vegetation change, and land use. Chaput and Gajewski (2016), Crema (2022), and Crema and Bevan (2021) support the discussion of how radiocarbon dates can, and cannot, inform histories of human population.

Reviewer #3 (Remarks to the Author)

I work on Quaternary paleoclimate reconstructions and with Earth system scientists, and can comment on the relevance and impact of this study. It is indeed assumed by many that humans have played a significant and increasing role in shaping Holocene vegetation assemblage dynamics for the 8.5 to 2 ka BP study interval. This assumption is relevant, for instance, to the vigorous debate about whether the Anthropocene should be viewed as having gradually unfolded over tens of millennia (the “Anthropocene event” of Gibbard et al., 2022; see also the gradualist depiction in fig. 1 of Ellis, 2023), or whether significant impacts are restricted to more recent

times. The conclusion that humans were not responsible for broad-scale ecological processes during this interval of study is therefore important to this debate, and it sets a baseline for human-driven impacts that were to later overwhelm Earth system functioning.

The study itself is based on the novel statistical analysis of two very large and independent data sets. The conclusions seem to be well supported by the data, and careful thought is given to explaining their limitations. There is one controversial aspect of Holocene climate that is not addressed but perhaps could be. It concerns claims by Ruddiman et al. (e.g. 2003, 2007, 2020) that the emergence of intensive rice cultivation in eastern Asia was responsible for the release of significant carbon dioxide and methane into the atmosphere, from 7 ka and 5 ka respectively. This would explain why global temperatures remain relatively flat even as summer insolation at 65°N declined from its peak in the Early Holocene. These claims have been questioned by Zalasiewicz et al. (2024) who noted that isotopic and modelling studies are consistent with oceanic outgassing rather than anthropogenic causes for these rises in greenhouse gases. Moreover, Summerhayes et al. (2024) noted that global average insolation may have in fact risen slightly over the past 7 kyr, potentially leading to the degassing of the global ocean which is by far the largest CO₂ reservoir on the planet. The question then, as to whether there is evidence within the pollen record for substantial cultivation of rice in eastern Asia, becomes relevant in assessing the extent of human impacts on the Earth system.

Response. We thank the reviewer for placing the study within the debate over early human influence on the Earth system. The history of rice cultivation in eastern Asia is clearly relevant to the early anthropogenic hypothesis. Our evidence, however, cannot resolve that question. The ten pollen assemblage properties describe broad changes in richness, evenness, and compositional reorganisation; they do not identify rice pollen or reconstruct the extent and intensity of cultivation. Nor do the present models examine greenhouse-gas sources. Consequently, the larger estimated climatic contribution in our analyses neither supports nor refutes the proposal that early rice agriculture affected atmospheric carbon dioxide or methane. A direct test would require harmonised taxon-level rice evidence, regional histories of cultivation and land use, and an analytical design linking those histories to greenhouse-gas change.

Changes in the revised article. The final Discussion paragraph now locates the result within the early anthropogenic debate while defining its evidential boundary. It explains why broad pollen assemblage properties cannot substitute for taxon-level pollen evidence of rice cultivation, archaeological histories of cultivation, or evidence concerning the sources of Holocene greenhouse gases.

I consider the MS to be publishable after minor revision, and perhaps following consideration of the question of eastern Asia rice cultivation. The Holocene is crucial to our understanding of the rise of civilizations and the emergence of the modern world. This MS will have a profound impact on the way we approach these fundamental issues.

Response. We thank the reviewer for this assessment. The revised article now discusses eastern-Asian rice cultivation as an important historical and Earth-system question while making clear that the present broad-scale pollen analysis cannot adjudicate between competing explanations of Holocene greenhouse-gas change.

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Response. We thank the reviewer for this literature. It is now used to frame the early anthropogenic debate and the alternative anthropogenic and oceanic explanations for Holocene greenhouse-gas change. We state explicitly that the evidence analysed here does not distinguish between those mechanisms.

Minor detailed comments on the MS

Line 69. Should there be consistency here in the precision, i.e. 8.5–2.0 ka BP? See Line 336 where this consistency is maintained.

Response. Agreed. We now use 8.5–2.0 ka BP consistently when specifying the study interval.

Line 82. For land use, reference to Our World in Data would be better: <https://ourworldindata.org/land-use>.

Response. Thank you for the suggestion. We added the Our World in Data synthesis to document the long-term expansion of cropland, grazing land, and settlements. Peer-reviewed studies remain the basis for causal interpretations and for the geological discussion of the Anthropocene.

Line 95. not “middle Holocene” but “Middle Holocene” if you are referring to the formal substage/subepoch (8.2–4.2 ka), or “mid-Holocene” if more loosely.

Response. Corrected. We use “Middle Holocene” for the formal subdivision and “mid-Holocene” for informal usage.

Line 113. Pleistocene and Holocene epochs (not periods).

Response. Corrected to “epochs.”

Line 116. “the term “period” not needed here because the Holocene can only refer to a unit of time.

Response. Corrected. We removed the unnecessary use of “period” with Holocene.

Line 124. reflects, not reflect (“each” refers to the singular).

Response. Corrected to “reflects.”

Line 134. “on the period between 8.5 and 2.0 ka BP” if the consistency in precision is justified.

Response. Corrected, with the study interval consistently expressed as 8.5–2.0 ka BP or “between 8.5 and 2.0 ka BP,” according to sentence structure.

Line 147. we would expect an altered covariance between the PAPs within different climate zones on continental regions.

Response. Corrected and clarified. H2 is now expressed as a prediction about historical changes in the relationships among pollen assemblage properties within continental groups and their regions, rather than as a claim about the response of particular plant taxa.

Line 150. This subheading is a little confusing because while it references H1, it describes the result rather than the stated hypothesis. I realize that restating the hypothesis could be

misleading here, so perhaps frame the hypothesis as “H1: Holocene palaeoclimate versus human impact”.

Response. Agreed. We revised the subheading to “H1: Holocene human impact secondary to palaeoclimate,” thereby linking the hypothesis to its ecological and temporal setting while indicating the observed direction of the result.

Line 157. Middle to Late Holocene – you are referring to official subseries of the Holocene, so the first initial of the modifiers “middle” and “late” must be capitalized. If this is not done, you will need to define what will then be informal terms.

Response. Corrected to “Middle to Late Holocene” where the formal subdivisions are intended.

Line 175. Again an issue of consistency in precision. In Asia 3.5 ka BP so for Europe should that be 3.0 ka BP?

Response. Corrected. Ages are now reported with consistent precision within each comparison.

Line 218. palaeoclimate data between 8.5 and 2.0 ka BP to investigate ... (then delete “8.5 and 2.0 ka BP” at the end of the sentence).

Response. Corrected and the duplicated interval was removed.

Line 220. Our first hypothesis posited that ...

Response. Corrected. The sentence now begins with an explicit formulation of the first hypothesis.

Line 241. PAPs not PAPS.

Response. Corrected throughout to “PAPs.”

Line 275. “mid” is not a stand-alone word in most dictionaries, so it would be “mid-Holocene” as an informal interval of time.

Response. Corrected to “mid-Holocene.”

Line 326. “factors like” better as “such factors as”.

Response. Corrected to “such factors as.”

Line 336. “from 8.5 to 2.0 ka BP”. Use an en-dash when referring to, but not specifying, a range, such as “over the range 8.5–2.0 ka BP”. I note here that “2.0 ka unlike elsewhere is states to two significant figures, which seems appropriate precision if supported by the dates themselves.

Response. Corrected. We use “from 8.5 to 2.0 ka BP” in prose and an en dash in the adjectival or compact range “8.5–2.0 ka BP.”

Line 467. You do make it clear that Eric C Grimm is no longer with us, but are you perhaps

dedicating this MS to the memory of the person rather than the person themselves? Remember to change “manuscript” to “article” when revising the MS.

Response. Corrected. The article is now dedicated to the memory of Eric C. Grimm, and submission-facing prose refers to the revised work as an article.

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4. Bhatta, K. P. et al. Exploring spatio-temporal patterns of palynological changes in Asia during the Holocene. *Frontiers in Ecology and Evolution* 11, (2023).
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